

INSTALLATION & OPERATIONS MANUAL

EXCELSIOR ALTERNATIVE FUELS

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GENERAL INSTRUCTIONS

This manual is a guide for installation, operation and maintenance of your new Schutte-Buffalo Hammer Mill.

Please note that we properly lubricate all bearings at the factory at final assembly. We then test run each piece of equipment at normal operating speeds checking for proper fits, correct alignments, bearing temperatures, free running and, of course, vibration.

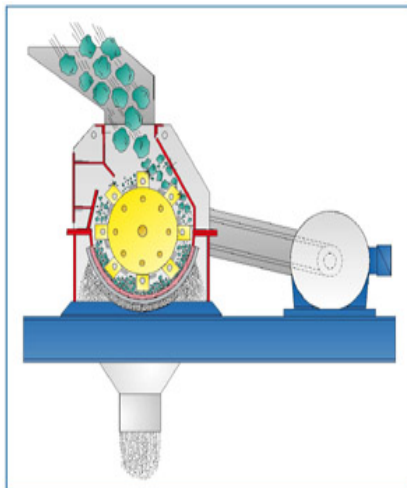
Whenever practical, we completely assemble each machine and its components including motor and drive and ship as a unit.

Given proper installation, correct lubrication and maintenance, your Schutte-Buffalo Hammer Mill will give you many years of trouble free service.

Simplicity, safety and ease of operation are the outstanding features of your *SCHUTTE* Hammer Mill. Like any other machine, its productivity and life depends upon proper installation and a conscientious program of preventative maintenance.

SERIAL NUMBER AND MODEL NUMBER

In any correspondence or parts orders pertaining to your Schutte-Buffer Hammer Mill please, specify model and serial number, as this will help us to give you better service and avoid any possible confusion.



MODEL NUMBER: 1320

SERIAL NUMBER: S07-12-003

INSTALLATION

Three things should be considered when locating the mill.

- Nearness to collector
- Convenience for feeding
- Foundation

PLACEMENT: Place the collector high enough so the ground product will discharge and flow by gravity through whatever distribution system is installed beneath the collector. The mill's blow line should run vertically to the necessary height, using only one 90-degree long sweep elbow into the collector inlet and a minimum of horizontal run. Any additional elbows and bends create excessive backpressure, reducing the grinding capacity. The *SCHUTTE* fan system will elevate materials 100 ft. total vertical and horizontal run.

You will notice that your *SCHUTTE* mill is specially designed so that any feeding attachment can be permanently mounted, as the mill hood is hinged to allow it to be opened without disturbing the attachment. The feed table and throat of the mill should be open, allowing as much air as possible to enter with the product. By *boxing in* the throat, the fan system will starve for air and not function properly.

NOTE: Above two paragraphs do not apply to gravity discharge units.

FOUNDATION: Support your mill on concrete or steel. The setting must be heavy and strong enough to carry the vibration and weight of the mill and its driving motor. This foundation can be isolated from any building foundations so as not to transmit vibration into the building. The foundation should be level and uniformly even with anchor bolts located as per the dimensions on certified Schutte-Buffalo assembly prints. Bolt the mill solidly, *shimming* to make certain that it remains level in both directions. Exercise care when drawing up the foundation bolts so the mill frame is not warped due to an uneven or unlevelled foundation. Although a Schutte-Buffalo Hammermill is a heavily built machine, care should be exercised to prevent damage in handling.

FEED CHUTE: All mills should be equipped with proper feed chutes or hoods. These should not only provide proper feeding but prevent kickback of material as well.

ACCESSIBILITY: Hammers, hammer rods, screens and/or screen bars are replaced, without removing the mill top, through covers in the mill. To change the shafts or discs, the mill top must be opened. Provide sufficient working space around your mill so that the mill top and Hammermill rods can be removed. The hammer rods are removed from the side opposite the drive on most mills.

ELECTRICAL: After fastening the mill, wire the motor starters and any other devices you may utilize in your setup. The wiring should be connected according to the wiring diagrams in your starter and standard electrical procedures. We supply wiring diagrams with all pre-wired panels ordered from us.

OPERATION

STARTING: Before starting the mill, inspect the grinding chamber carefully and turn the mill by hand to be certain that it is free of all objects that may have found their way into the interior of the mill. Check the direction of rotation to see that it is the same as indicated by the arrows on the sides of the mill or on our assembly drawing. When you start the mill, let it obtain full speed before feeding any material into it. Lastly, let the mill empty itself completely before shutting down.

FEEDING: Material fed into a Schutte-Buffalo Hammer Mill should be spread evenly over the entire feed width and should be conveyed into the mill continuously and uniformly at a rate not greater than what the mill will handle. Bear in mind that surge feeding is inefficient, as it tends to overload the mill thereby causing the machine to use excess power and possibly stall it. An alternate overload and under load, condition of operation will average far below the results obtained by steady feeding of the right amount of material. For a smooth and even flow of material, we recommend the use of an automatic feeder.

Feed size of material should be limited to the maximum size recommended by the factory. Excessively large pieces of material can make the hammers deflect seriously; causing excessive wear in the hammer holes and hammer rods. Uncrushable materials such as steel and other tramp metal should be removed from the feed before they enter the mill even though the mill may be equipped with a metal trap (optional). A metal trap is by no means 100% effective and severe damage may occur before the uncrushable material can be removed.

DISCHARGE: The chute, air system, belt, screw conveyor or other conveying system used to move the ground product must have enough capacity to handle peak loads without allowing any accumulation of product inside the mill. This is necessary to prevent plugging the mill and shutting down its operation.

FINENESS: The fineness of the finished product is determined by a combination of screen size, mill speed and number of hammers. Changing any one of these three items will change the fineness of the finished product. The combination of factors used on your mill, as sent from the factory, was determined by actual testing of your material or by previous experience with your material. If, for some reason, you desire to change the fineness of your finished product, you can easily do so by one of the following methods. The methods are in the order of their importance with the most effective method first. Do not change more than one thing at a time so there is no confusion as to which change is producing the results.

CHANGING SCREEN SIZE: The easiest, most practical and most effective method to change fineness of the grind is to change the size of the perforations in the screen or the opening size in the screen bar. A screen with smaller holes will produce a finer product because it will retain the particles in the grinding chamber longer and conversely a screen with larger holes will produce a coarser product.

CHANGING SPEED: Increasing the speed of the mill increases the number of hammer blows per minute. In addition, the increase in speed means the hammers hit harder blows, thus increasing the shattering effect. Increasing the speed, therefore, produces a finer product.

NUMBER OF HAMMERS: Increasing the number of hammers also increases the number of hammer blows per minute, therefore producing a finer product. Extreme care must be taken in changing the number of hammers so that the balance of the rotor is maintained.

MAINTENANCE

Any machine that has rotating or moving parts requires periodic maintenance. This is especially true of size reduction equipment, since any machine that reduces the size of a material also gradually wears away the grinding elements. Although Schutte-Buffalo hammer mills are extremely rugged, the following guidelines should be followed.

INSPECTIONS: We recommend that you make periodic inspections of your Schutte Buffalo Hammer Mill. These inspections should include hammers, hammer rods, discs, screen, other wearing parts and belt tension. Also, make sure that all bolts are securely tightened.

SPARE PARTS: It is best to stock replacement parts to avoid interruption of your production. We recommend that you stock a minimum of one set of hammers, one set of hammer rods and one set of screens and/or screen bars. A spare set of bearings is also a good investment.

LUBRICATION: The frequency of re-lubrication depends upon the actual operating conditions. Dusty or wet conditions may require lubrication that is more frequent. Avoid over greasing high-speed machines to prevent excessive bearing heating. We suggest that you contact your local representative of the bearing manufacturer for answers to any questions you may have.

V-BELT DRIVES: Your mill is equipped with a compact drive, Dodge Dyna –V or equal. V-Belts should be checked periodically for belt wear and for proper tension. When properly tensioned, the belts will have only a slight bow on the slack side when operating under load. If the belt slips, check for proper tensioning or for oil on the belts. When adjusting belts, make sure the mill is not running. Replace belts when worn using the same compact type belts. Belts should only be replaced in matched sets. Care should be taken when adjusting or changing belts so that the faces of the sheaves are kept parallel and in line. Improperly aligned sheaves will cause overheating and belt failure.

ALIGNMENT FOR DIRECT CONNECTION: Because of handling during shipping and installation of direct connected units, the coupling may become misaligned and must be realigned according to the manufacturers' instruction sheet packed with the coupling. A perfect alignment increases the life expectancy of both mill and motor bearings and eliminates excessive vibration.

HAMMERS: The standard hammers used in our mills are double end reversible types, which have four wearing edges. When hammers become dull to the point where the machine is not grinding efficiently or when the hammers are well rounded, turn the hammers so a new sharp edge will be in service. One row of hammers should be turned at a time so the rotor balance will not be changed. Do not allow the hammers to wear too far or you will lose the opposite two cutting edges. Hammers can be changed through the access door in the mill top with the hammer rods sliding through the access doors on the sides of the mill top, or through the main access door.

The hammers installed in your mill were carefully balanced at the factory. New sets of replacement hammers sent from the factory are also carefully balanced and wire in rows. Always work on one row of hammers at a time and do not interchange hammers from one row to another. Careless handling of hammers can cause excessive vibration.

HAMMER RODS: When changing hammers, inspect the hammer rods for wear, which shows as grooving on the rods. If the rods are deeply grooved, replace them. Hammer rods are also carefully balanced and should be replaced only as balanced sets.

SCREEN: Your mill is equipped with either a perforated metal screen or screen bars. Screens and screen bars are interchangeable. Replace the mill screen or screen bars when the edges of the holes or bars are well rounded due to wear. Screens can be removed and replaced through the access doors provided on one side of the mill base below the bearings, or through the main access door.

On some “B” Series Hammer Mills, the side entry screen feature is not available. In these mills, the screens are changed by opening the mill top and inserting the screen in the slots provided in the mill base.

Note – Not all mills are equipped with the following.

FAN HOUSING LINER: Do not let the fan housing liner plate wear through before replacing it. The liner protects the fan housing and should be inspected periodically to determine whether it should be replaced.

FAN ROTOR: The fan rotor should also be inspected periodically to determine the amount of wear. To replace, simply remove fan-housing cover, which gives access to the fan rotor.

WEAR PLATES: Wear plates protect the mill housing. Mills that are equipped with replaceable wear plates should be inspected periodically to determine the amount of wear. Do not let the wear plates wear through before replacing them.

TROUBLESHOOTING

VIBRATION AND NOISE: After a mill has been in service for some time, wear may affect the balance of the rotor assembly causing vibration and noise. Another source of noise may be the V-belts which can slip due to lack of tension or which may be worn causing a flapping sound. If the mill is repeatedly overloaded by feeding oversize or uncrushable material, the hammers will deflect (or lay back) causing excessive wear. This wear elongates the holes and reduces the rod diameters, permitting the hammers to hit the screen or other parts. Change hammers and rods immediately and avoid improper feeding.

LOSS OF CAPACITY OR PRODUCT SIZE: This problem is largely due to wear of the grinding elements and can be corrected by reversing the hammers and changing the screen or screen bars, if worn.

CLOGGING: Clogging can be caused by feeding material into the mill too fast or not conveying product away from the mill fast enough. It can also be caused by worn parts that are not grinding properly or due to excessive moisture, oil, etc. in the feed to the hammer mill.

SAFETY INSTRUCTIONS

We urge that this Hammermill be operated and equipped in strict accordance with all safety regulations.

Do not neglect your responsibility to see that this Hammermill is operated, maintained, and equipped with total regard to operator safety. No one can be responsible for neglect but you.

Study this manual *BEFORE* attempting to run the Hammermill.

Never operate or place the Hammermill under power with any guard or cover open or removed.

Be alert for loose, worn or broken parts. Report all of these items and any unusual noise or action of the machine to your supervisor.

Keep your Hammermill properly maintained.

Never wear loose clothing or jewelry, which could become entangled in the Hammermill system.

Never lay tools or material on the Hammermill where they might interfere with the operation of the unit.

Be alert whenever operating the Hammermill.

NEVER REACH INTO THE BREAKING AREA OF THE HAMMERMILL WHILE IT IS OPERATING. INSURE THAT THE ENTIRE UNIT IS SHUT OFF BEFORE ATTEMPTING TO REACH INTO THE BREAKING AREA.

Never operate the Hammermill on a material it was not designed to break.

Consult the manufacturer for recommended application of your Hammermill.

Never reach across moving parts.

All materials and lifting devices should be of sufficient capacity to handle weight of the Hammermill, refer to general arrangement drawing for correct weights.

Turn main electrical power off and lock out power when changing hammers/cutters, tightening components or accomplishing any maintenance work.

All guards should be replaced before mill is put back into operation after removal for any reason.

The actual breaking operation can produce extreme forces. Ensure that hoppers are securely fastened to Hammermill and that no personnel are in the Hammermill vicinity during shredding unless adequate safeguards are taken to protect personnel from moving objects.

High voltage is used to power this Hammermill. Only authorized electricians should correct any electrical problem.

It is impossible to foresee all possible uses, misuses, and application of this machine. Your best protection against injuries is to use common sense and to think before attempting any questionable uses of this machine.

INSTALLING INSTRUCTIONS FOR HAMMERS

BALANCE BY THE CLOCK METHOD

Upon arrival, note that each row or stack of hammers are numbered on the top hammer. They are numbered 1-8, or 1-10 depending on how many rows the rotor employs.

TO INSTALL:

Start with the row marked "1". As illustrated in Figure #1, insert one hammer between every other disc. (In some cases, two hammers will be placed between each disc, depending on style of rotor).

Row 2 will follow with the hammers being staggered between the discs opposite the previous row. (See Figure #2)

Continue installing the hammers in this manner working around the rotor in numerical order. (See Figure #1)

Each hammer has four (4) wearing edges. Hammers should be rotated when the grinding edge wears to approximately half way between the hole and the original corner. Do not allow hammers to wear beyond this point or you will lose the ability to utilize all four edges. (See Figure #3)

It is recommended that you replace the hammers rods when they show signs of wear (general rule of thumb - one set of rods following the usage of all four edges of the hammer). When rotating the hammers, check for premature wear to the rod itself. Excessive grooving of the rod will indicate a need for replacement.

Install collars or cotter pins in the end of each rod following hammer rotation or replacement.

INSTALLATION PROCEDURE FOR MIDDLE ASSEMBLY OR BEARING REPLACEMENT

1. Clean off bearing housing support pads
2. Clean and/or debur bearing housing base
3. Install assembly in mill with bearing housing mounting bolts loose
4. "Seat" bearing housing on support pads by hitting on top with a mallet
5. Align housings parallel to edge of support pad
6. Center rotor in mill frame maintaining an equal distance between end discs and interior sidewalls.
7. Set proper hammer to screen clearance – approximately 3/8"
8. Tighten mounting bolts: one at a time in each housing while observing lift or gap, that is created between the housing and support pad on the un-tighten side of the same housing. Shim to eliminate this lift or gap and bolt down. Repeat procedure on other housing.

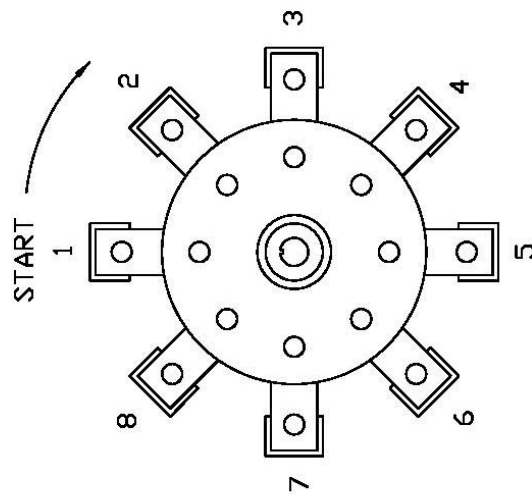


FIGURE 3

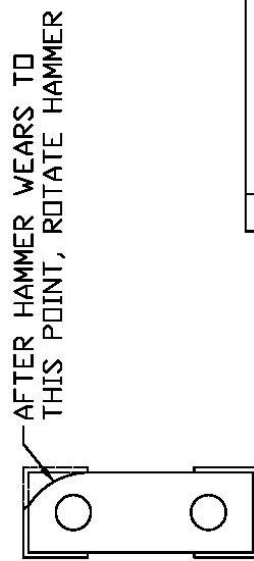
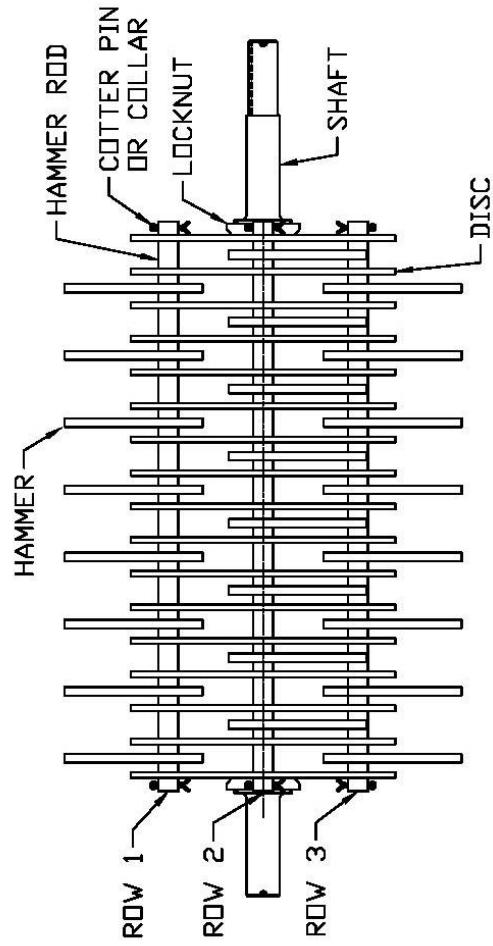
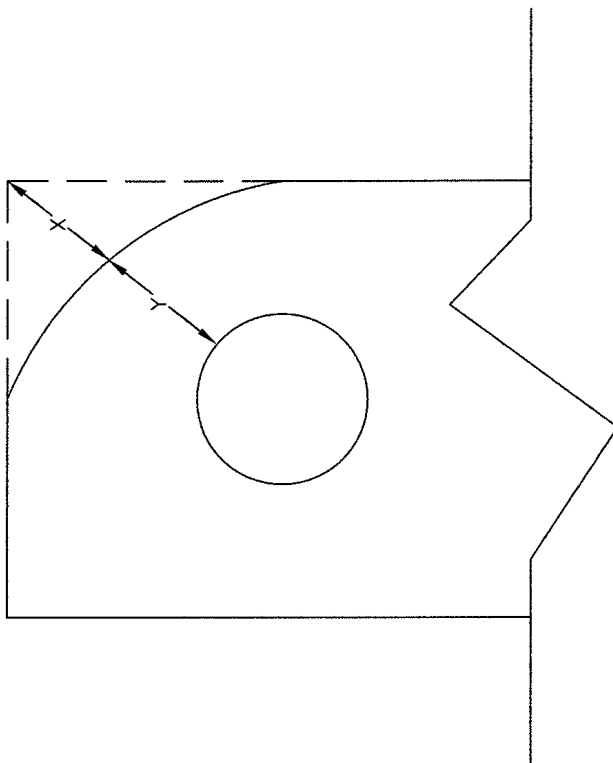


FIGURE 2

[illegible]

ROTATE HAMMERS WHEN DISTANCE X=DISTANCE Y



LETTER		ALTERNATE		DATE	BY	Schutte- Buffalo 81 Depot Street Buffalo, N.Y. 14203		DESIG. XXX DATE X/X/XX DESIG. X-X DATE X-X	DRAWN BY CHECKED IN CHARGE
TITLE		80661A HALL HAMMER ROTATION		80661A HALL HAMMER ROTATION		80661A HALL HAMMER ROTATION		80661A HALL HAMMER ROTATION	

ORDERING PARTS

We give you three easy ways to order:

- Fax your order
- Call us
- E-mail: sales@hammermills.com

Schutte-Buffalo Hammer Mill, LLC

61 Depot Street
Buffalo, NY 14206

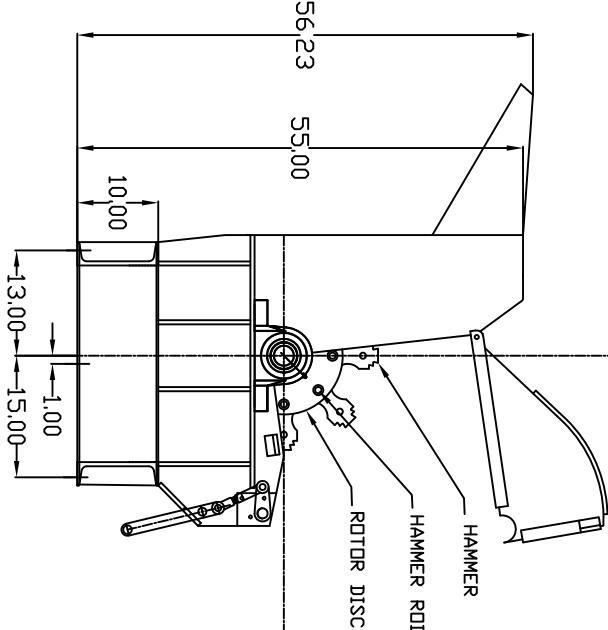
Phone: 716-855-1555
Toll Free: 1-800-447-4634
Fax: 716-855-3417

www.hammermills.com

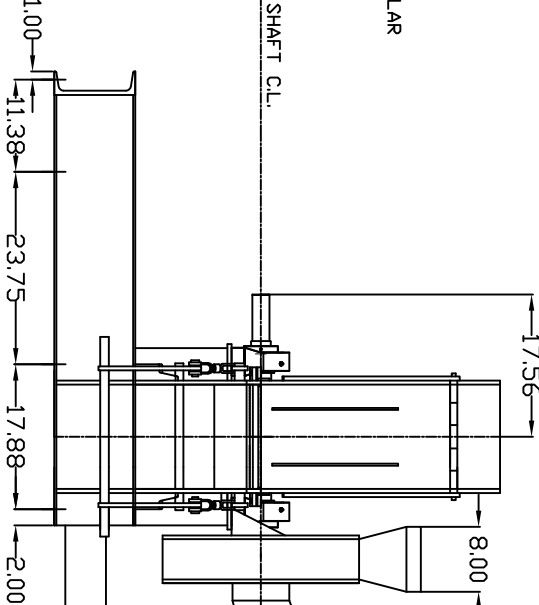
NOTE: ALL BASE MOUNTING HOLES 1.03"

MOTOR NOT SHOWN
IN THIS VIEW

SHAFT
C.L.

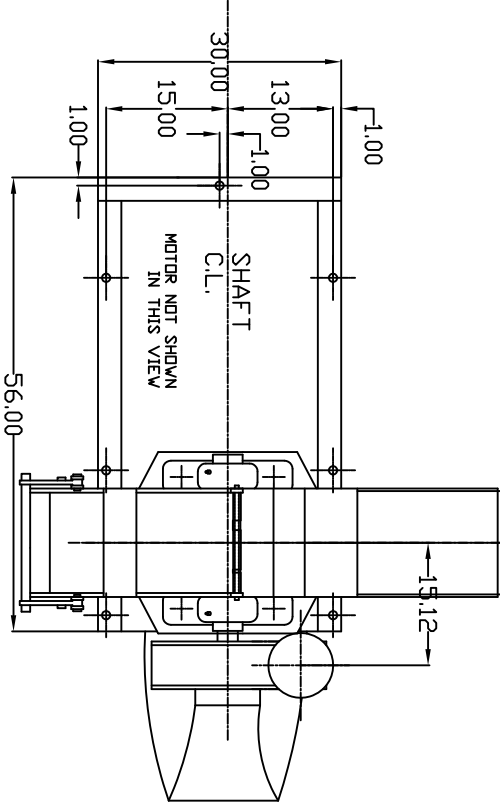


SHAFT C.L.



FEED OPENING

INLET
C.L.

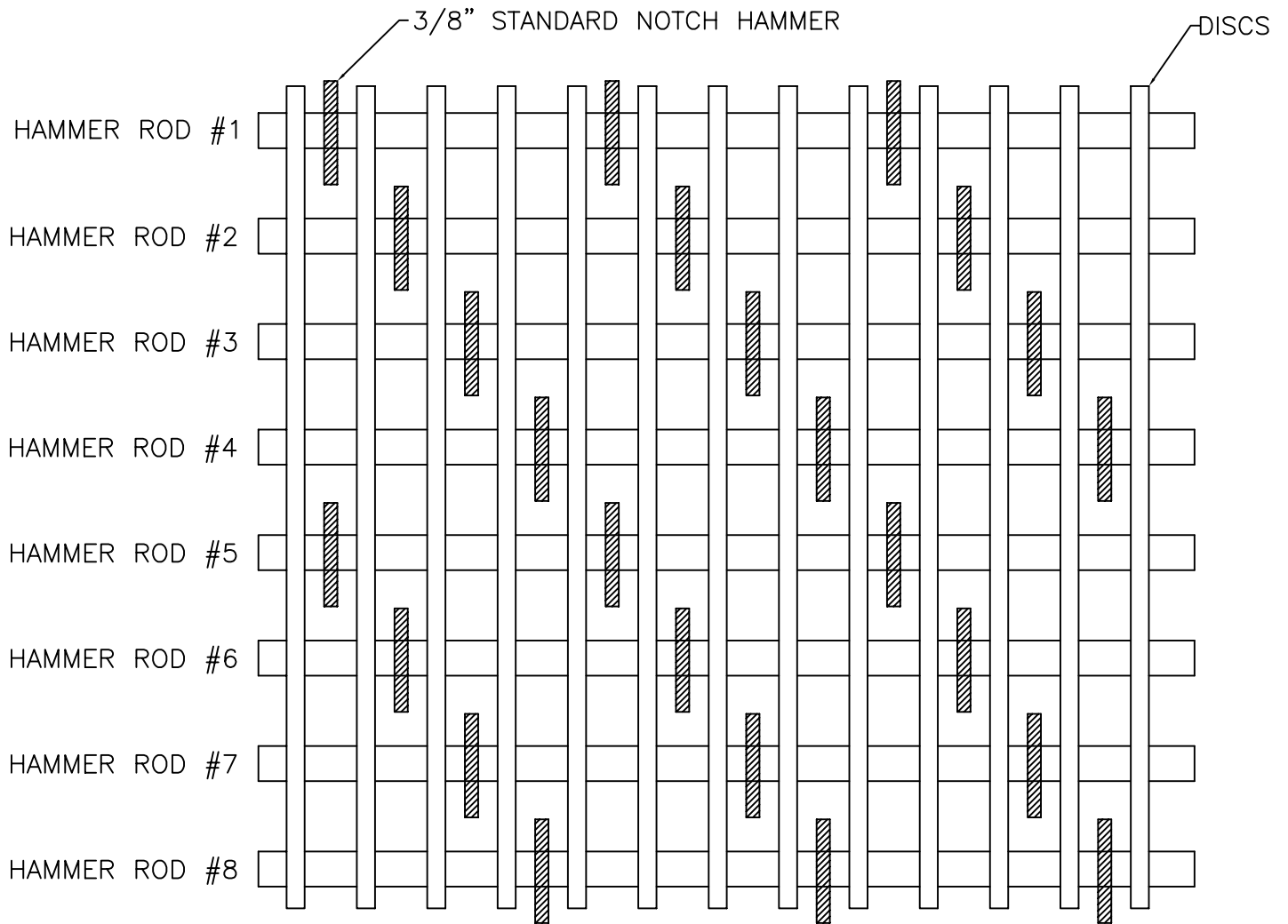


UNLESS SHOWN OTHERWISE TOLERANCES			
1) .001	2) .005	3) .010	4) .020
5) .030	6) .040	7) .050	8) .060
9) .070	10) .080	11) .090	12) .100
13) .125	14) .150	15) .175	16) .200
17) .250	18) .300	19) .375	20) .450
21) .500	22) .600	23) .750	24) .900
25) 1.000	26) 1.250	27) 1.500	28) 1.750
29) 2.000	30) 2.500	31) 3.000	32) 3.500
33) 4.000	34) 4.500	35) 5.000	36) 5.500
37) 6.000	38) 6.500	39) 7.000	40) 7.500
41) 8.000	42) 8.500	43) 9.000	44) 9.500
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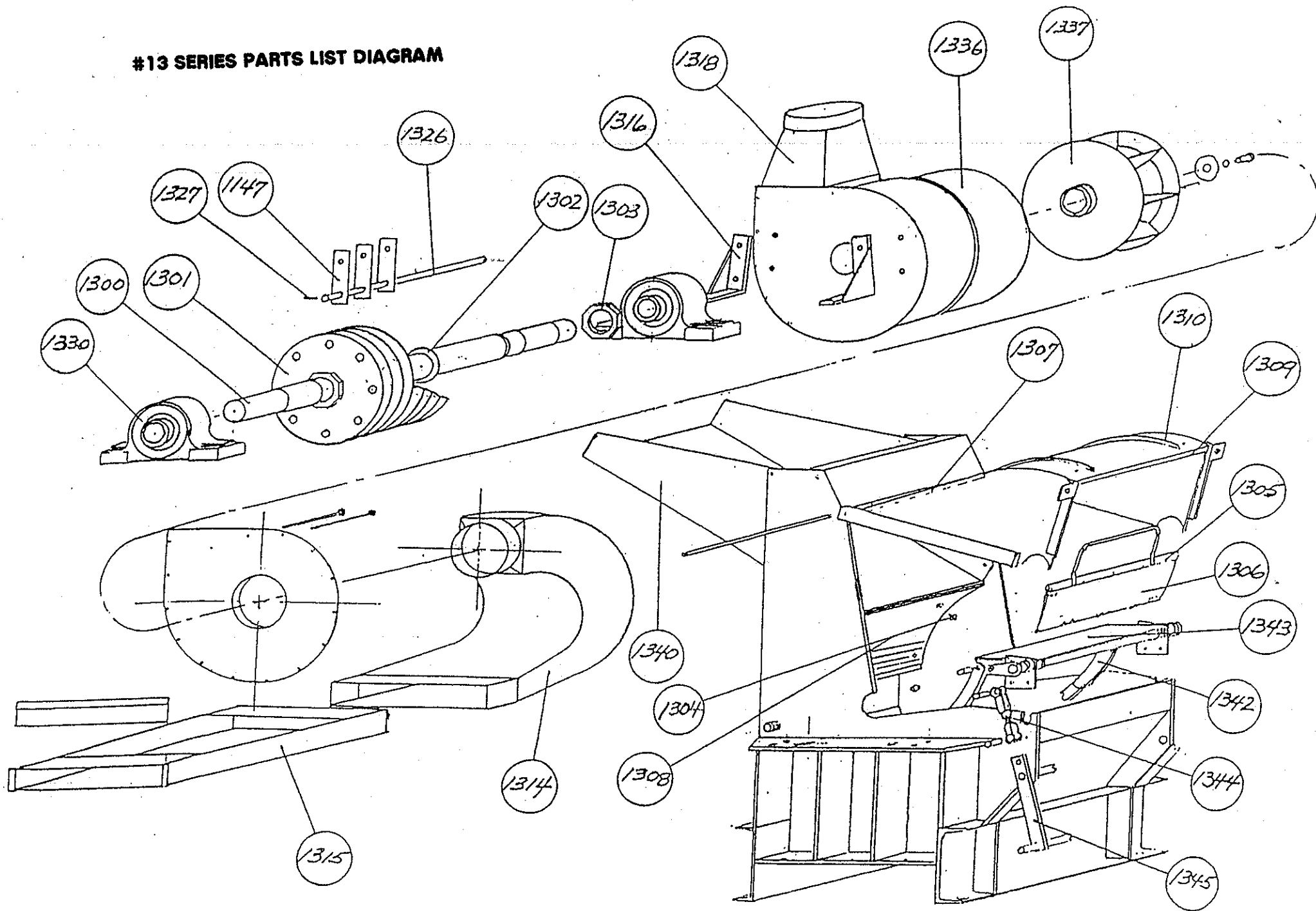
MODEL 1020 OR 1320

HAMMER ARRANGEMENT

24 HAMMERS — 12 SPACES — 13 DISCS



#13 SERIES PARTS LIST DIAGRAM



#13 SERIES PARTS LIST

DIAGRAM – KEY

1300	Shaft
1301	Disc, Round 1 / 4 Gauge
1302	Spacer
1303	Disc Nut (Right Or Left Hand)
1304	Liner Plate Bolt
1305	Screen Bar With Handle
1306	Screen
1307	Upper Rear Liner Plate
1308	Front Ribbed Liner Plate
1309	Rear Ribbed Liner Plate
1310	Rear Hood
1314	Suction Pipe
1315	Bottom Pan With Slide Control
1316	Fan Housing Bracket (Right Or Left Hand)
1318	Fan Housing
1326	Hammer Rod (3/4" Or 1" Diameter)
1327	Collars
1330	Bearing With Housing
1336	Fan Liner
1337	Fan Rotor
1340	Feed Table, Heavy Gauge
1342	Bottom Linkage
1343	Swing Locking Plate
1344	Adjustable Yoke
1345	Operating Lever
1150	Hammer

Link-Belt® BEARING DIVISION

Series 200, 300, 3-U200, 3-Y200,
3-W200, U200, Y200, W200,
U300 and Y300

Link-Belt® Ball Bearing Units Service Instructions B-BBU-22

IMPORTANT — Read Carefully

These instructions are provided to aid in the proper installation, operation and maintenance of Link-Belt Series 200, 300, 3-U200, 3-Y200, 3-W200, U200, Y200, W200, U300 and Y300 ball bearing units. They should be carefully read and followed. Failure to do so may result in unsatisfactory service as well as serious personal injury or property damage.

U200 series ball bearing units up to and including 2 1/4 inch shaft size are secured to the shaft with two setscrews threaded into the inner ring. Larger size U200 series and all U300 series units are equipped with spring locking collars having two setscrews. All W200, Y200, and Y300 series ball bearing units feature eccentric locking collars with one setscrew.

CAUTION

The reliability built in all Link-Belt bearings can be realized in service only when they are correctly selected, properly installed, protected and maintained.

The correct selection of bearings or mounted units requires that the magnitude and nature of all loads, speeds, alignment, mounting, operating requirements and maintenance be adequately considered. The selection of materials for and design of housings, shafting, fasteners, seals and accessories as well as provisions for installation and maintenance must follow good engineering principles.

When axial (thrust) loads are present, shaft collars or other means may be required to transmit the thrust from the shaft to the inner ring and prevent the bearing from slipping axially on the shaft.

Housings must be selected and installed with regard to the degree and direction of the forces that will occur. Housings should not be used under tension loads except with adequate safety factors. For this reason pillow blocks are best suited to withstand radial loads passing through the base. When heavy loads or shock loads are possible, it is most important to mount a unit so that the line of force passes directly into its base, or so that the unit is directly and substantially supported other than through its mounting bolts. Where the line of force falls outside the base, such as with horizontal or uplift loads on pillow blocks, serious housing and fastener deflection or failure may occur. These conditions may require designs using different materials, fasteners, mounting design, stop bars, etc., together with proper safety factors. When these conditions are unavoidable, Link-Belt Bearing Division should be consulted.

The following points of installation and operation are very important:

- Cleanliness** — Keep dirt, water, and metal chips off all parts.
- Careful Handling** — Hammer blows or improper use of force can damage precision parts.
- Shaft Fits** — Bearings should have proper fits on the shafts to minimize fret wear. See installation instructions for shaft tolerances. When mounting bearings on a used or worn shaft, care must be taken to clean up the shaft journal and rebuild, as necessary, to the required tolerances. Never replace bearings on a shaft which is bent or which has been damaged or softened by a torch.

D. Bolts — Housing mounting bolt tightness is important to prevent the housing from shifting, and to adequately support loads.

E. Collars and Setscrews — Eccentric cam locking collars must be securely tightened in the direction of shaft rotation with the setscrew locked, and setscrews in spring locking collars or in setscrew mounted bearings must be properly torqued to prevent the shaft from slipping in the inner ring and to prevent loosening during operation.

F. Free Rotation and Alignment — Check for free rotation before machine start-up to assure that final alignment is proper. Although bearings are alignable in housings, they will not "self-align" in many cases and destructive moment loads may be imposed if initial installation alignment is not provided.

G. Lubrication — Units designed for relubrication must be adequately lubricated. A bearing not properly lubricated can run to destruction and possibly cause damage to other components. Add grease slowly. Rapid application may blow the seals and allow the grease to escape.

H. Temperature — Certain components of these units are made from materials which have temperature limits. Unless units are specifically made for high temperature applications, operating temperatures must not exceed 225°F for units with E, E2, N, U, or housing felt seals, or 275°F for units with H seals.

INSTALLATION

- Check Shaft** — Shafting must be clean, round, straight, free of burrs and nicks and of correct size.

For most operating conditions with low to moderate loads and speeds, the following shaft tolerances are recommended:

Shaft Diameter	Recommended Tolerance
1/2" thru 2"	Nominal to minus .0005"
2 1/8" thru 4"	Nominal to minus .0010"

Satisfactory performance may be obtained with increased shaft tolerances under some less severe conditions. Very difficult applications may require a light interference fit of the bearing on the shaft. Consult Link-Belt Bearing Division for recommendations.

- Shaft Preparation** — When frequent removal of bearings is anticipated, file flats on the shaft approximately 1/16" deep under the setscrew locations. This provides for easy removal, as bearings will clear the burrs caused by correctly tightened setscrews.

- Lubricate Shaft & Bearing Bore** — Coat the shaft and bearing bore with grease or oil to facilitate assembly.

- Assemble on Shaft** — If it is necessary to tap or press units on the shaft, use a hardwood block, soft steel bar or tube against the end of the inner ring. DO NOT strike or exert pressure on housing or seals.

Setscrew Mounted Units — Back out all setscrews to clear the shaft and slide the units into position on the shaft.

Spring Locking Collar Units — Back out all collar setscrews to clear the shaft and slide the units into position on the shaft.

Eccentric Cam Locking Collar Units — Slide each unit into position on the shaft with the cam locking collar positioned for engagement with the inner ring cam.

- Preliminary Unit Location** — With both bearing units on the shaft, preliminarily locate the shaft in position. Lightly bolt the units to the mounting structure. SAE Grade 5 mounting bolts can be used. Bridge over housing mounting bolt slots with heavy flat washers. The use of heavy spring lockwashers under bolt heads and nuts is good practice to prevent loosening.

- Final Positioning** — Establish the final shaft position. Align the bearing units by hand, if required, or use a shaft or a rubber mallet. Securely bolt both

TABLE 1 — Inner Ring and Collar Setscrew Tightening Torque

Shaft Size (in.)	Seating Torque (in-lbs.)				
	W200	Y200	U200	U300	Y300
1/2 to 1 1/16	87 - 92	87 - 92	33 - 40	33 - 40	
3/4	87 - 92	87 - 92	87 - 92	33 - 40	
1 1/16 to 1	87 - 92	87 - 92	87 - 92	87 - 92	
1 - 1/16 to 1 - 3/16	165 - 185	165 - 185	87 - 92	165 - 185	290 - 325
1 - 1/4 to 1 - 7/16	290 - 325	290 - 325	165 - 185	165 - 185	290 - 325
1 - 1/2 to 1 - 9/16	290 - 325	290 - 325	165 - 185	165 - 185	
1 - 5/8 to 1 - 3/4	290 - 325	290 - 325	165 - 185	290 - 325	430 - 460
1 - 13/16 to 1 - 15/16	290 - 325	290 - 325	290 - 325	290 - 325	620 - 680
2 to 2 - 3/16		430 - 460	290 - 325	430 - 460	620 - 680
2 - 1/4 to 2 - 7/16		430 - 460	290 - 325	430 - 460	620 - 680
2 - 1/2 to 2 - 11/16			430 - 460	620 - 680	1225 - 1350
2 - 3/4 to 2 - 15/16			620 - 680	620 - 680	2125 - 2350
3 to 3 - 1/2			620 - 680	1225 - 1350	
3 - 9/16 to 3 - 3/4			1225 - 1350	1225 - 1350	
3 - 13/16 to 4			1225 - 1350	2125 - 2350	

Note: Satisfactory tightness may be obtained by using a standard hex wrench (50-52 Minimum RHN C) and tightening until a slight permanent twist is obtained in wrench for each setscrew.

Link-Belt® BEARING DIVISION

TABLE 2 — Grease Lubrication

OPERATING ENVIRONMENT		Bearing Operating Temperature	Greasing Interval (1)	Use Grease Equivalent to these (2)
Dirt Exposure	Moisture Exposure			
Slight	None	32°F to 120°F 120° to 160°F 160°F to 200°F	6 months 2-4 months 1-2 months	High quality NLGI No. 2 multi-purpose ball bearing greases are generally satisfactory. Consultation with a reputable lubricant supplier is recommended.
Moderate to Heavy		32°F to 160°F 160°F to 200°F	1-4 weeks 1 week	
Slight to Heavy	Direct water splash or exposure to outdoor environment	32°F to 200°F	Daily to 1 week or as determined by inspection of installation	Mobil Oil Corp., Mobilith AW 2 Shell Oil Co., Alvania EP 2 Texaco Inc., Premium RB 2
Slight	None	-60°F to 32°F	Determined by inspection of installation	Shell Oil Co., Aeroshell 7A Mobil Oil Corp., Mobiltemp SHC 32
		above 200°F		Special lubrication and seals may be required.

(1) Frequency of regreasing will vary, depending on the hours of operation, temperature and surrounding conditions.

(2) Do not use diester ester type lubricants without consulting Link-Belt Bearing Division, which can provide assistance in selection of lubricants given knowledge of operating conditions.

Link-Belt Bearing Division cannot be held responsible for performance of individual batches of grease. Changes in lubricant specifications, performance, and lubricant guarantees are the responsibility of the lubricant manufacturer.

LIMITED WARRANTY — LIABILITY

A. IT IS EXPRESSLY AGREED THAT THE FOLLOWING WARRANTY IS GIVEN IN LIEU OF ALL OTHER WARRANTIES, WHETHER EXPRESS, IMPLIED OR STATUTORY, INCLUDING THOSE OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, AND OF ANY OTHER OBLIGATION OR LIABILITY ON OUR PART OF ANY KIND OR NATURE WHATSOEVER.

No representative of ours has any authority to waive, alter, vary or add to the terms hereof without prior approval in writing, to our customer, signed by an officer of our company. It is expressly agreed that the entire warranty given to the customer is embodied in this writing; that this writing constitutes the final expression of the parties' agreement with respect to warranties; and that it is complete and exclusive statement of the terms of the warranty.

We warrant to our customers that all Products manufactured by us will be free from defects in material and workmanship at the time of shipment to our customer for a period of one (1) year from the date of shipment. All warranty claims must be submitted to us within ten days of discovery of defects within the warranty period, or shall be deemed waived. As to Products or parts thereof that are proven to have been defective at the time of shipment, and that were not damaged in shipment, the sole and exclusive remedy shall be repair or replacement of the defective parts or repayment of the proportionate purchase price for such Products or parts, at our option. Replacement parts shall be shipped free of charge f.o.b. our factory.

This warranty shall not apply to any Product which has been subject to misuse; misapplication, neglect (including but not limited to improper maintenance and storage); accident; improper installation, modification (including but not limited to use of unauthorized parts or attachments), adjustment, repair or lubrication. Misuse also includes, without implied limitation, deterioration in the Product or part caused by chemical action, wear caused by the presence of abrasive materials, and improper lubrication. Identifiable items manufactured by others but installed in or affixed to our Products are not warranted by us but, bear only those warranties, express or implied, given by the manufacturer of that item, if any.

Responsibility for system design to insure proper use and application of Link-Belt Products within their published specifications and ratings rests solely with customer. This includes without implied limitation analysis of loads created by torsional vibrations within the entire system regardless of how induced.

B. It is expressly agreed that our liability for any damages arising out of or related to this transaction, or the use of our Products, whether in contract or in tort, is limited to the repair or replacement of the Products, or the parts thereof by us, or to a refund of the proportionate purchase price. We will not be liable for any other injury, loss, damage or expense, whether direct or consequential, including but not limited to loss of use, income, profit, production, or increased cost of operation, or spoilage of or damage to material, arising in connection with the sale, installation, use of, inability to use, or the replacement of, or late delivery of, our Products.

C. It is also expressly agreed that any cause of action for breach of any warranty must be brought within one year from the date of the breach.

Link-Belt Bearing Division
Rexnord Corporation
7601 Rockville Road, Box 85, Indianapolis, Indiana 46206

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Stock Carrying Distributors and Representatives Throughout the World

units to the mounting structure by correctly torquing the bolts.

- Locking The Bearings** — Before the collars or setscrews are tightened, it should be possible to axially move the shaft by hand thru the bearings bores. If this cannot be done, repeat step 6. When satisfied, lock one bearing to the shaft. If one unit is an expansion type, lock the fixed bearing first.

Setscrew Mounted And Spring Locking Collar Units - Tighten both setscrews to the seating torque values in table 1, or, using a good quality individual hex wrench, tighten each setscrew until a slight permanent twist is attained in the shank of the wrench.

Eccentric Cam Locking Collar Units - Engage the collar on the inner ring cam. Rotate the collar in the direction of shaft rotation until it becomes tight using a hammer and a drift pin, continue tightening until solid. Tighten the setscrew.

Rotate the shaft a few revolutions to center the bearings. Locate expansion units in the center of their axial movement. Tighten the setscrews or collars on the remaining unit using the same method.

- Tabbed End Closure Installation** - For YG300 & UG300 only — Insert the end closure tabs into the housing loading slots. Tap on the end closure face by hand or with a rubber mallet until the end closure seats against the housing. Fill the gap between the end closure outer diameter and housing bore with a bead of silicone sealant/adhesive.

- Check for Noise, Freedom of Rotation Or Vibration** — Any condition of strain, irregular rotational torque, abnormal sound or vibration should be investigated. Look for improper alignment, out-of-balance, bent shaft, distorted or weak supports, broken welds, etc. Installation should be rechecked and corrected as required.

The use of stop bars welded to the support at each end of units leaving about 1/32" space after units are securely fastened. These bars assure proper location of the units and prevent shifting when loads are applied.

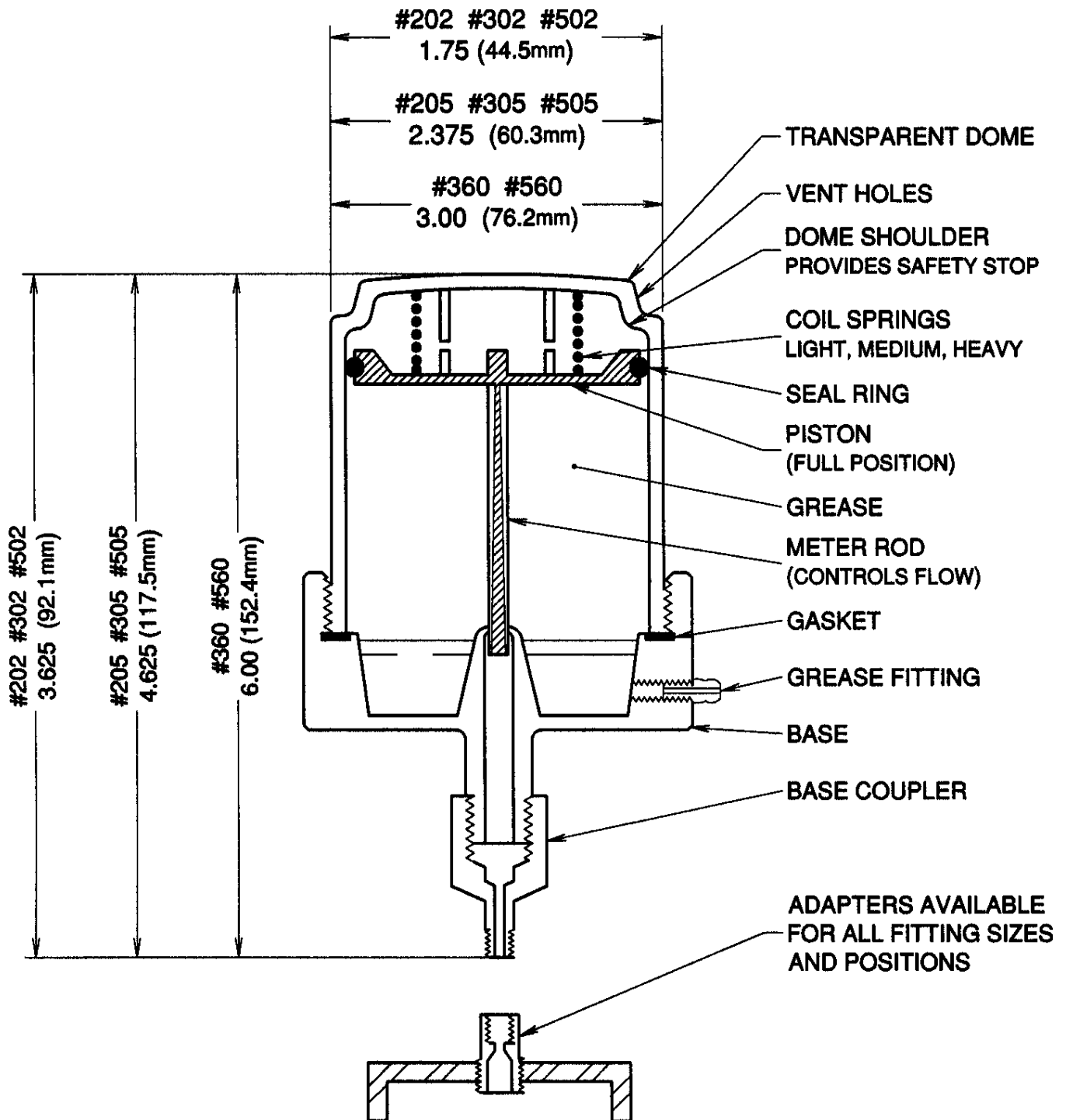
LUBRICATION

Units are prelubricated. With relubricatable designs no additional lubricant is required for startup. As a precaution, if equipment is to be built and left idle for any period of time prior to actual use relubricatable bearings should be filled 100% full to provide maximum protection from corrosion, etc. The suggested relubrication schedule under Table 2 is a general guide.

The specific conditions on an application such as exact hours of operation, temperature, moisture, speed and dirt govern the required lubrication cycle. This can be determined by inspection of the flushed out lubricant during a trial period of operation. Add grease slowly. Use a sufficient volume of grease to purge the bearing seals of old lubricant. It is preferable to rotate bearings during relubrication where good safety practice permits. Inspection of bearing installations at least every six months is recommended. Any unusual noise or vibration change should be immediately investigated.

PRODUCT INFORMATION SHEET

GREASE FEEDERS - INSTALLATION DIMENSIONS



Illustrations and specifications are not binding in detail. Designs are subject to modification and improvement without notice.



LUBESITE SYSTEMS, INC.

A SUBSIDIARY OF VESPER CORPORATION

Tel: 1-252-522-4688 • Toll Free: 1-800-526-5211 • Fax: 1-252-522-2913

E-Mail: ici1368@icomnet.com • Website: www.lubbsite.com

LubeSite® Grease Feeder 200 - 700 Series Installation Instructions

FOR BEST RESULTS, FOLLOW THESE INSTRUCTIONS:

IMPORTANT

BEFORE MOUNTING

- 1 USE PROPER SPRING** -- Every LubeSite unit comes factory-equipped with MEDIUM SPRING (SILVER color) which gives correct grease feeding pressure for most applications. A lightweight spring (BLUE) and a heavyweight spring (RED) are available. Consult the Spring Selection Guide Chart for selection to suit your application.
- 2 REMOVE GREASE FITTING FROM EQUIPMENT OR BEARING HOUSING** -- Check to see that the thread on the grease fitting is a 1/8" PT. You may need another LubeSite adapter to mount the LubeSite unit (refer to accessories).
- 3 BEFORE MOUNTING, FILL** LubeSite with a non-separating grease by connecting a grease gun to the side grease fitting. Fill with grease until the grease comes out of the base coupler. First filling may be above Caution line to assure no air pockets. Outlet must be unobstructed as over-pressuring can result in top separating from base which can cause serious head or eye injury. Mount LubeSite on equipment immediately.

MOUNT

- 4 SCREW THE LUBESITE ASSEMBLY DIRECTLY INTO THE GREASE FITTING HOLE -- TURN HAND TIGHT ONLY.** Mounting 360 and 560 Models in any position other than vertical may require a support brace.

REFILL

- 5 LUBESITE CAN BE REFILLED WHILE MOUNTED ON EQUIPMENT.** This should be accomplished before the seal ring is down on the top of the base. Simply refill by attaching the grease gun to the grease fitting and fill until the seal ring rises to the bottom edge of the "Caution" label. **DO NOT OVERFILL!** Overfilling can cause over-pressure which can result in top separation from base which can cause serious head or eye injury. For refilling hard to reach applications, use the LubeLine Remote Refill Line.

Adherence to these instructions will give you many years of LubeSite trouble-free automatic lubrication.

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E-Mail: ici1368@icomnet.com • Website: www.lubesite.com

Metal-bodied limit switches

ENM 2

Standard switch according to DIN EN 50041
Protection class IP 65



- Increased number of switching functions for safety applications
- Galvanically separated contacts
- Forced disconnection of NC contacts (type dependant)
- Person-protection function ⊕ (depending on model)
- Protection class IP 65
- Aluminum body and lid
- Operating device: 4 x 90° actuator positions
- Cable entry M 20 x 1.5
- International approvals

Mounting

- 2 x M 5 screws, adjustment via elongated holes
- 2 x M 5 screws for safety applications without additional attachment

Mounting advantages

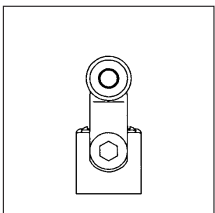
- Increased wiring space
- Earthing surface on same level as switching system
- Turret
Lever: 7.5° / 15° adjustable
- Contact made independent of direction
- Screw terminals with self lifting clamps for easy wiring
- Fine adjustment of switching point via adjusting screw
- Captive lid screws

Options

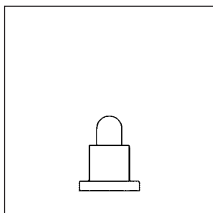
- Latching with contact locking (strain lock-release for safety reasons)
- LED indication of switching status
- EEx certificated versions available acc. ATEX 100 a guideline 94/9/EG
⊕ II 2 G EEx d II C T6
TÜV 03 ATEX 2043 X



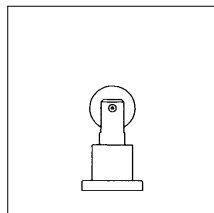
Standard actuating element DIN EN 50041



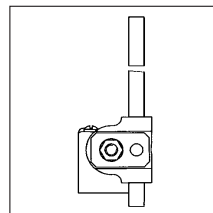
Form A



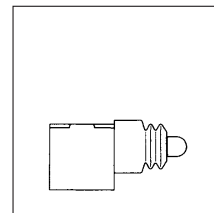
Form B



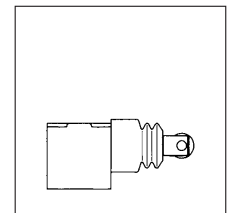
Form C



Form D



Form F
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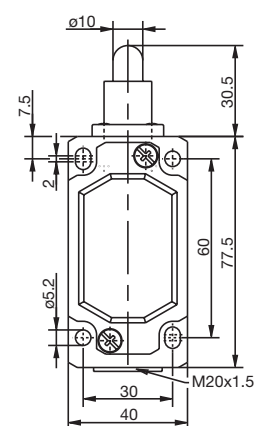
Form G
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request)

Standard contact configuration

Switching element	Switch function	Switch contact	Designation	Voltage	Continuous current
Slow-action	changeover	1NC/1NO	U1/U1Z	400 V	10 A
Snap-action	changeover	1NC/1NO	SU1/SU1Z	400 V	10 A
Slow-action	normally-closed	2NC	A2/A2Z	400 V	10 A
Snap-action	normally-closed	2NC	SA2/SA2Z	250 V	10 A
Slow-action (overlapping)	changeover	1NC/1NO	UV1/UV1Z	400 V	10 A
Slow-action (overlapping)	changeover	2NC/1NO	UV15/UV15Z	400 V	5 A

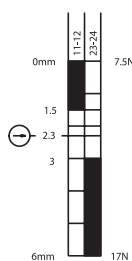
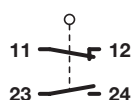
Optional contact configuration

Switching element	Switch function	Switch contact	Designation	Voltage	Continuous current
Slow-action	normally-open	2NO	E2	250 V	10 A
Snap-action	normally-open	2NO	SE2	250 V	10 A
Slow-action	changeover	2NC/1NO	U15/U15Z	400 V	5 A
Slow-action	changeover	1NC/2NO	U16/U16Z	400 V	5 A
Slow-action (overlapping)	changeover	1NC/2NO	UV16/UV16Z	400 V	5 A

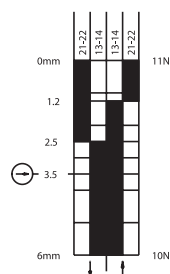
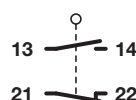


Contact travel diagrams and graphic symbols

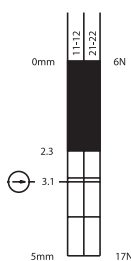
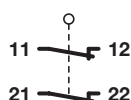
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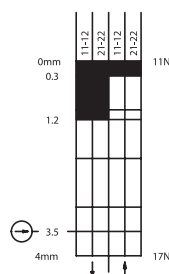
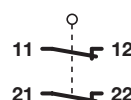
SU1Z



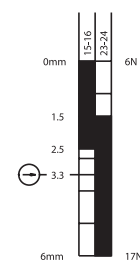
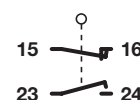
A2Z



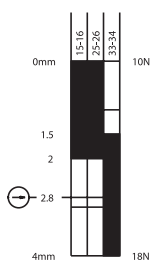
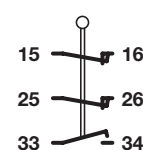
SA2Z



UV1Z

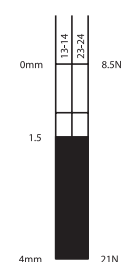
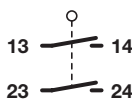


UV15Z

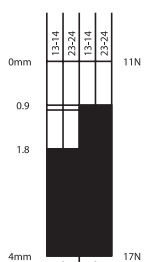
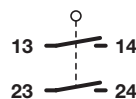


On request

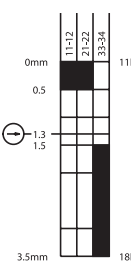
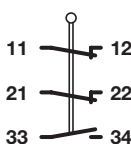
E2



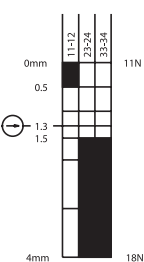
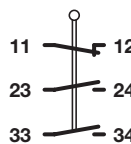
SE2



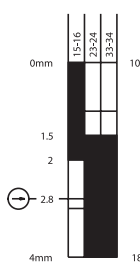
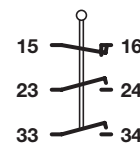
U15Z



U16Z



UV16Z





Designation

Part number

Switching diagram

➔ Positive break according to IEC 947-5-1 Chap. 3

Za: changeover contact is not galvanically isolated

Zb: changeover contact is galvanically isolated

Slow-action contact / snap-action contact

Gasket inside (w/w/outside (w))

ENM2-U1Z iw	ENM2-SU1Z iw	ENM2-A2Z iw	ENM2-SA2Z iw	ENM2-E2 iw	ENM2-SE2 iw
608.7102.001	608.7352.002	608.7802.003	608.7852.020	608.7802.021	608.7852.022
●/-	-/●	●/-	-/●	●/-	-/●
iw	iw	iw	iw	iw	iw

Contact travel mm Tol. ± 0.25 mm		Actuating force N Tol. $\pm 10\%$
Switch angle degree Tol. $\pm 3.5^\circ$		Actuating torque Ncm Tol. $\pm 10\%$
		
	In Out	

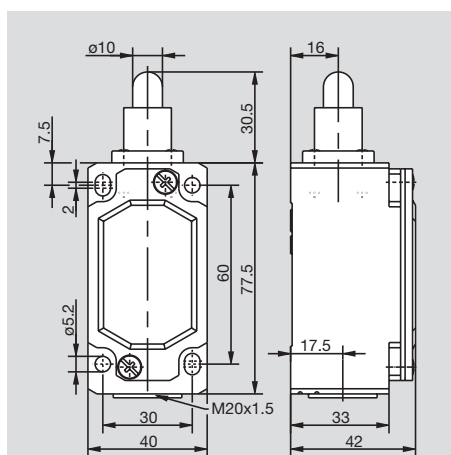
Voltage	max.
Continuous current	max.
Making current, acc. to IEC 947-5-1 AC 15/DC 13	
Switching frequency	max.
Mech. operational life – number of switching cycles	
Ambient temperature	min./max.

400 V AC	400 V AC	400 V AC	250 V AC	250 V AC	250 V AC
10 A	10 A	10 A	10 A	10 A	10 A
●	●	●	●	●	●
100/min.	100/min.	100/min.	100/min.	100/min.	100/min.
10 x 10 ⁶	10 x 10 ⁶	10 x 10 ⁶	3 x 10 ⁶	3 x 10 ⁶	3 x 10 ⁶
-30 °C/+80 °C	-30 °C/+80 °C	-30 °C/+80 °C	-30 °C/+80 °C	-30 °C/+80 °C	-30 °C/+80 °C
-22 °F/+176 °F	-22 °F/+176 °F	-22 °F/+176 °F	-22 °F/+176 °F	-22 °F/+176 °F	-22 °F/+176 °F

Standard actuating element	Form
Approvals (applied for)	
Weight	
Delivery: ex-stock/built to order	

B	B	B	C	C	C
UL, CSA	UL, CSA	UL, CSA	UL, CSA	UL, CSA	UL, CSA
0.20 kg	0.20 kg	0.20 kg	0.20 kg	0.20 kg	0.20 kg
●/—	●/—	●/—	—/●	●/—	—/●

All dimensions in mm



A & A -Plate Magnets-

A & A Plate Magnets are used to remove scrap metal from items such as grains, glass, chemicals, plastics, minerals and foodstuffs. These units are primarily used in ducts and chutes or suspended over conveyor belts. Plate magnets prevent damage to equipment and product contamination by effectively removing tramp iron from free-flowing materials and materials under pressure. They provide permanent magnetic strength and are easily installed to fit most granulators in use today. Five standard models are available with widths starting at 6 inches. Price quotations are available on request, as are optional features:

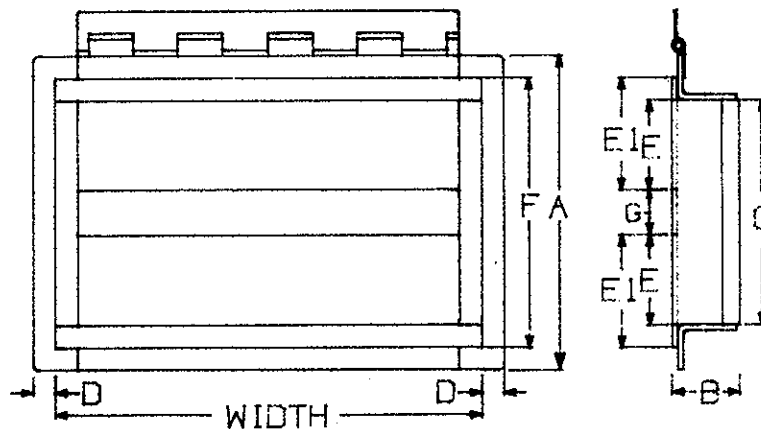
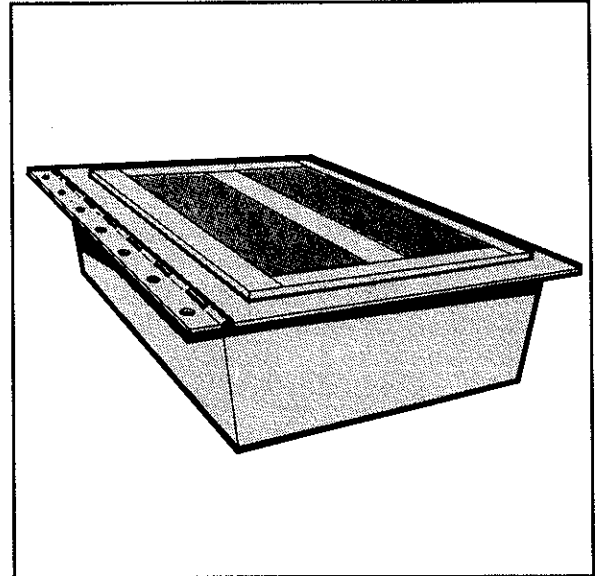
- **All Stainless Steel Construction** for resistance to a corrosive atmosphere and for materials often required for the food handling and processing industries.

- **Hinge Mounting** for installation in chutes and ducts. Magnet swings down for easy cleaning.

- **Special Barriers & Steps** for more positive removal of tramp iron.

- **Liquid Tight Construction** prevents moisture from entering unit.

- **Special Mounting Holes** either tapped holes in back face or through holes in flanges for stationary installation. (Flange will be increased by 1/2" to allow for holes.)



Model	Effective Product Depth	DIMENSION IN INCHES						
		A	B	C	D	E	F	G
PM 150	2	7	1-1/2	5	1/2	2	6	1
PM 250	3	9	1-1/2	7	1/2	3	8	1
PM 350	4	9	2-5/8	7	1/2	3	8	1
PM 450	5	11-1/2	3-13/16	9-1/2	1/2	4	10-1/2	1-1/2
PM 650	6-7	16-7/8	6-5/8	14-1/2	3/4	*6-3/8	15-1/4	2-1/2

Width according to customer specifications.

*E1 - On 650 Model only pole piece extends to flange

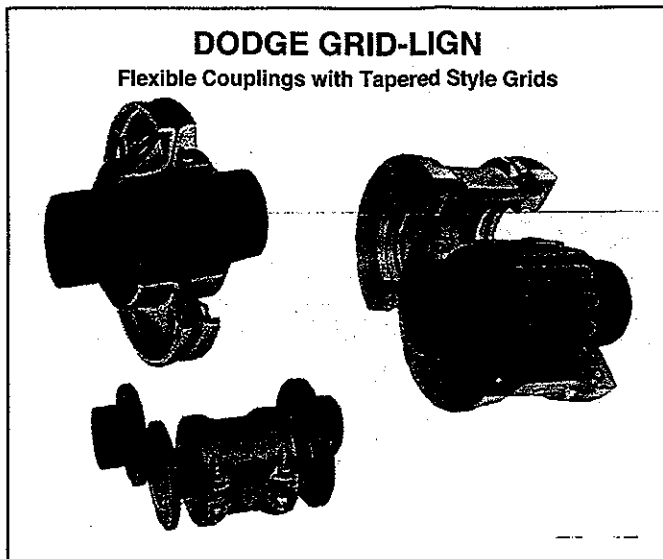
A & A Magnetics, Inc.

P.O. Box 1427
Woodstock, IL 60098

Phone: (815) 338-6054
FAX: (815) 338-8590

INSTRUCTION MANUAL FOR DODGE® GRID-LIGN® FLEXIBLE COUPLINGS

Sizes 1020-1140 T10 (Close Coupled, Horizontally Split Cover)
T20 (Close coupled, Vertically Split Covers)
T31 (Full Spacer Type)
T35 (Half Spacer Type)



STEP 1: PREASSEMBLY INSPECTION

All parts should be examined for any damage during the shipping and handling process. Measurement should be taken to ensure parts meet applications requirements, such as hub and shaft fits, shaft separation, etc. All parts must be clean and free of any foreign material before attempting assembly.

STEP 2: INSTALLATION OF KEYS

Install keys in respective shafts. Keys should fit keyseat with a tight fit on the sides and slight clearance over the key. Use plastic oil sealing compound around keys to prevent loss of lubricant.

STEP 3: MOUNTING HUBS

Fit hubs with ends of shaft per the following recommendations:

A. Interference Fit with a straight bore hub requires the hub be expanded prior to mounting. Heat hub in oil bath or oven until bore is larger than shaft diameter. Refer to Chart 1 for temperature guidelines. Use extreme caution if heating bore with an open flame: To avoid distortion, it is essential that hub be heated evenly, not exceeding +450°F.

NOTE: The interference fit limitation for a keyed hub is a maximum .00075" interference per inch of shaft diameter.

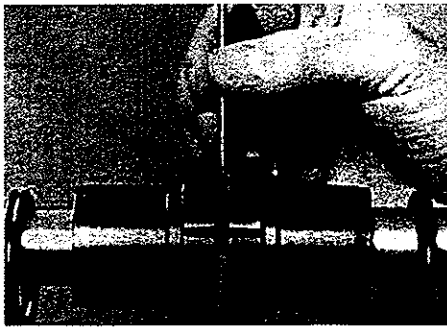
B. Clearance Fit with a straight bore hub is recommended only for light duty applications. Slip hub onto shaft and tighten set screws to value listed in Table 8. Set screws should be checked periodically for tightness.

C. Taper Bored hubs do not require preheating for assembly. Mount hub and align on shaft before drawing up on shaft the required distance. In most applications, the hub face will project beyond the small end of the shaft taper. Lock hub in place with proper locking device provided with shaft.

D. TAPER-LOCK® hubs are available in T10, T20, T31 and T35 couplings for light and medium applications. Install per instructions packed with the TAPER-LOCK bushings.

E. Shaft Separation for T10 and T20 couplings should be set per appropriate "G" dimensions in Table 1. For T31 and T35 spacer couplings, refer to Tables 6 and 7 for shaft separation dimensions.

WARNING: Because of the possible danger to persons(s) or property from accidents which may result from the improper use of products, it is important that correct procedures be followed. Products must be used in accordance with the engineering information specified in the catalog. Proper installation, maintenance and operation procedures must be observed. The instructions in the instruction manuals must be followed. Inspections should be made as necessary to assure safe operation under prevailing conditions. Proper guards and other suitable safety devices or procedures as may be desirable or as may be specified in safety codes should be provided, and are neither provided by Reliance Industrial Company nor are the responsibility of Reliance Industrial Company. This unit and its associated equipment must be installed, adjusted and maintained by qualified personnel who are familiar with the construction and operation of all equipment in the system and the potential hazards involved. When risk to persons or property may be involved, a holding device must be an integral part of the driven equipment beyond the speed reducer output shaft.



Shaft Separation

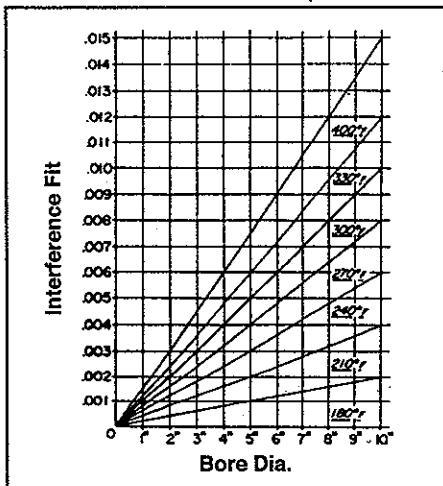


Grease Packing



Inserting Grid

Chart 1 — Temperature Chart



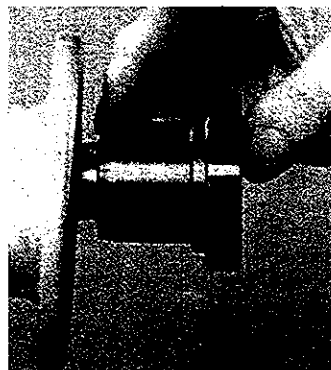
STEP 4: ASSEMBLY

4A: Assembly of T10 (Close coupled, horizontally split cover) coupling

- When mounting T10 couplings, lightly grease each of the two seals from the grid cover assembly. Place each seal far back on the respective shaft.
- After hubs are mounted, the couplings must be aligned according to Steps 5 and 6.
- Before inserting grid, hand pack hub teeth with lubricant. (Refer to Tables 1 and 2 for recommendations on type and quantity of lubricant.) Fit grid over hubs and starting at one end, work coils of grid between the teeth. Seat with a soft mallet. If grids are supplied in more than one segment, install so that all cut ends extend in the same direction.



T10 Hub Mounting



T20 Hub Mounting

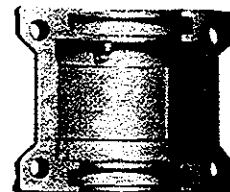


Figure 1
Anti-Rotation Lug

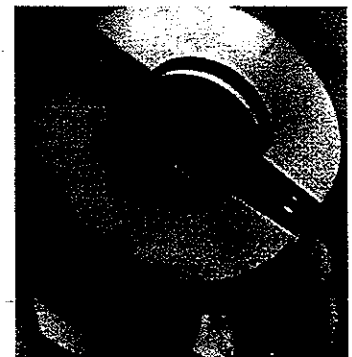


Figure 2
Cover Match Marks

Hand pack more lubricant around the grid and between the spaces of the grid after it is installed. Refer to Step 7 for important information on lubrication.

- Position the two cover seals on the hub and align with the grooves in the half cover. Place gaskets on flanges of bottom cover half. Assemble cover with match marks on same side. Install fasteners in cover halves with nut end of bolt nearest lube plug. (In this position, nuts are self retaining and do not require a wrench.) Tighten per T10 cover bolt tightening torque as shown in Table 1. (Sizes 1020-1090 have nut locking flats in cover.)

NOTE: T10 coupling may be mounted on vertical or inclined shafts. When doing so, assemble cover halves with anti-rotation lug and match mark up or on the higher side. (See Figures 1 and 2 below.)

4B: Assembly of T20 (Close coupled, vertically split cover) coupling

- Remove lube plugs from each cover half. Insert the two seal rings into each cover half. Lightly coat seals with grease. Place cover halves far back on respective shafts with flanges facing each other.
- After hubs are mounted, the coupling must be aligned according to Steps 5 and 6.
- Lubricate and install grid per Step 4 A-C for T10.
- Bring together the T20 vertical cover halves with gasket positioned between them, aligning all bolt holes. Lube holes should be 180° apart. Tighten bolts to tightening torque given in Table 1.

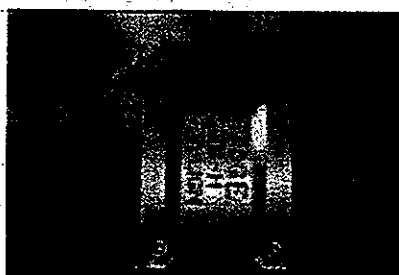
Table 1 — Recommended Operating Misalignment

Size	Misalignment Limits (1)		Coupling Gap "G"		BSE (Shaft Spacing) For DG-SP		BSE (Shaft Spacing) For DG-HSP		Spacer Flange Bolt Tightening Torque (lb.-in.)	Cover Bolt Tightening Torque For Type V, H, SP, HSP (lb.-in.)	Lubrication Weight (lbs.)
	Parallel (in.)	Angular (in.)	Type V, H, HSP (in.)	Type SP (in.)	Min. (in.)	Max. (in.)	Min. (in.)	Max. (in.)			
1020T	.005	.005	.125	.188	3.50	8.00	1.78	4.03	120	100	.06
1030T	.005	.005	.125	.188	3.50	8.50	1.78	4.28	120	100	.06
1040T	.005	.005	.125	.188	3.50	8.50	1.78	4.28	120	100	.12
1050T	.005	.005	.125	.188	4.38	8.50	2.22	4.28	250	200	.12
1060T	.010	.010	.125	.188	5.00	13.00	2.53	6.53	440	200	.25
1070T	.010	.010	.125	.188	5.00	13.00	2.53	6.53	440	200	.25
1080T	.010	.010	.125	.188	7.25	16.00	3.66	8.03	825	200	.38
1090T	.012	.012	.125	.188	7.25	16.00	3.66	8.03	1640	200	.56
1100T	.012	.012	.188	.250	8.00	16.00	4.06	8.06	2940	260	.94
1110T	.012	.012	.188	.250	8.25	16.00	4.19	8.06	2940	260	1.12
1120T	.012	.012	.250							650	1.60
1130T	.012	.012	.250							650	2.00
1140T	.012	.012	.250							650	2.50

(1) These are recommended installation misalignment limits. As operating misalignment increases, coupling life is reduced.



Rigid Hub Mounting



T31 Center Section Drops Out



Angular Alignment Feeler Gauge Method

4C: Assembly of T31 (Full Spacer) and T35 (Half Spacer)

- A. Install rigid shaft hubs per Step 3.
- B. The T31 and T35 seals and covers are mounted after the rigid shaft hubs are in place.
- C. Carefully stretch cover seals over teeth and onto spacer hub. Bolt each half spacer hub onto rigid shaft hubs and torque to specifications given in Table 1 (Spacer Flange Bolt Tightening Torque).
- D. Set shaft spacing per appropriate "G" dimensions per Table 1, 6 or 7.
- E. The coupling must be aligned according to Steps 5 and 6.
- F. Lubricate and install grid per Step 4 A-C for T10.
- G. Position cover seals on hubs and assemble cover with seals and gaskets per Step 4 D.
- H. To remove spacer with cover, loosen rigid hub bolts and compress spacer hubs to disengage pilots. Center section of spacer coupling will drop out without disturbing driver or driven shafts.

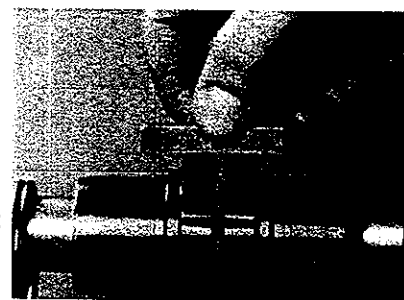
STEP 5: ANGULAR ALIGNMENT

NOTE: Proper alignment yields the longest service life.

- A. **Instrument Method** of checking alignment is recommended since it is most accurate. Rigidly attach dial

base to one of the hubs and indicator needle against a face of the other hub. Rotate both hubs 360°. Take indicator reading at four points, 90° apart. Adjust alignment until all four readings are within angular misalignment limits given in Table 1. To check alignment, relocate the dial base to the opposite hub and repeat the procedure.

- B. **Caliper Method** may be used if dial indicator is not available or shaft gap is too small; however, it is not the recommended method of checking alignment. Check with calipers or feeler gauge at four points, 90° apart. Adjust alignment until all four readings are within angular misalignment limits given in Table 1.



Parallel Alignment Straight Edge Method

STEP 6: PARALLEL ALIGNMENT

- A. **Instrument Method** is again recommended since it is most accurate. Rigidly attach dial base to one hub and set dial indicator button in contact with an outside diameter of opposite hub. Rotate both hubs 360°. Take indicator reading at four points, 90° apart.

Adjust alignment until all four readings are within parallel misalignment limits given in Table 1. To check alignment, relocate dial base on opposite hub and repeat the procedure. Recheck angular alignment.

B. A Straight Edge and Feeler Gauge may be used if a dial indicator is not available; however, it is not recommended as the most accurate method of checking alignment. Adjust alignment until straight edge appears to be resting squarely on both outside tooth diameters. Repeat procedure at 3 additional points, 90° apart. Refer to Table 1 for limits of parallel misalignment. Recheck angular alignment.

STEP 7: LUBRICATION

Remove both lube plugs in cover and insert lube fitting. Pump in appropriate lubricant (per Tables 1 and 2) until it is forced out the opposite lube hole, then install lube plugs in holes.

NOTE: All lube plugs must be installed before operating coupling.

Proper lubrication of all types of GRID-LIGN couplings is necessary for their efficient operation and long service life.

Table 2 lists lubricants, from several manufacturers, which will perform satisfactorily under average industrial conditions as long as the ambient temperature is within the limits of 0°F to 150°F (–180°C to +66°C). This table does not constitute a complete listing of acceptable lubricants, and it is not meant to prohibit the use of lubricants with equivalent properties. Lubricants required for severe operating conditions should be referred to the factory or a lubrication manufacturer's representative for recommendations. Under average industrial conditions it is recommended that GRID-LIGN coupling lubricants meet the specifications listed below.

GRID-LIGN Coupling Lubricant Recommendations:

- Ambient Temperature Range: 0°F to 150°F (–18°C to +66°C)
- NLGI Rating: 1 through 3 per ASTM D-217
- Minimum Base Oil Viscosity: 750 SSU @ 100°F per ASTM D-445
- Minimum Dropping Point: 220°F (105°C) per ASTM D-556 or D-2265
- Maximum Thickener Content: 8%
- Oxidation Resistance, Maximum Pressure Drop: 20 psi @ 100 hours per ASTM D-942
- Recommended Properties: anti-oxidation, anti-rust, extreme pressure

**Table 2 — Suggested Lubricants
(Ambient Temperature 0°F to 150°F)**

Manufacturer	Name of Lubricant
Amoco	Amolith #2
Citgo	Citgo HEP-2
Conoco	EP Conolith #2
Exxon	Ronex MP
Mobil	Mobilux EP-1 and EP-2
Shell	Alvania Grease #2
Sun Oil	Prestige 42

STEP 8: MAINTENANCE

Maintenance is recommended every 6 months to ensure long life. Disassemble coupling, remove old lubricant and clean. Visually inspect all parts. Replace any worn parts. Hand pack coupling with lubricant and repeat Step 7.

STEP 9: GRID REMOVAL

When it is necessary to disassemble coupling, remove cover halves. Beginning at a cut end of grid, carefully insert a screwdriver into loop. Using the teeth for leverage, gradually pry the grid up, alternating sides while working around the coupling.



Grid Removal

INTERCHANGE INFORMATION

The DODGE GRID-LIGN flexible couplings components are designed to be interchangeable with other **tapered grid** style coupling components. See Table 3 for more information.

**Table 3 — Tapered Grid Coupling
Nomenclature Interchange**

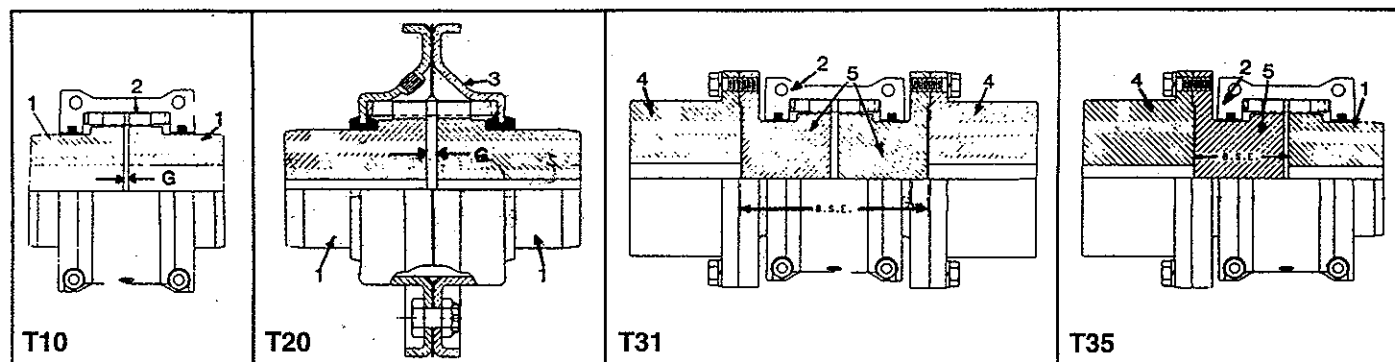
Coupling with Horizontally Split Covers			Coupling with Vertically Split Covers		
DODGE	Falk		DODGE	Falk	
1020T10	20T10	1020T10	1020T20	20T20	1020T20
1030T10	30T10	1030T10	1030T20	30T20	1030T20
1040T10	40T10	1040T10	1040T20	40T20	1040T20
1050T10	50T10	1050T10	1050T20	50T20	1050T20
1060T10	60T10	1060T10	1060T20	60T20	1060T20
1070T10	70T10	1070T10	1070T20	70T20	1070T20
1080T10	80T10	1080T10	1080T20	80T20	1080T20
1090T10	90T10	1090T10	1090T20	90T20	1090T20
1100T10	100T10	1100T10	1100T20	100T20	1100T20
1110T10	110T10	1110T10	1110T20	110T20	1110T20
1120T10	120T10	1120T10			
1130T10	130T10	1130T10			
1140T10	140T10	1140T10			
Spacer Coupling			Half Spacer Coupling		
DODGE	Falk		DODGE	Falk	
1020T31	20T31	1020T31	1020T35	20T35	1020T35
1030T31	30T31	1030T31	1030T35	30T35	1030T35
1040T31	40T31	1040T31	1040T35	40T35	1040T35
1050T31	50T31	1050T31	1050T35	50T35	1050T35
1060T31	60T31	1060T31	1060T35	60T35	1060T35
1070T31	70T31	1070T31	1070T35	70T35	1070T35
1080T31	80T31	1080T31	1080T35	80T35	1080T35
1090T31	90T31	1090T31	1090T35	90T35	1090T35
1100T31	100T31	1100T31	1100T35	100T35	1100T35
1110T31	110T31	1110T31	1110T35	110T35	1110T35

Table 4 – GRID-LIGN Coupling Components

Cplg. Type	Ref. No.	Name of Part	Part Number and Quantity Required per Coupling Size												
			1020T	1030T	1040T	1050T	1060T	1070T	1080T	1090T	1100T	1110T	1120T	1130T	1140T
Type T10	1	Std. Hub – RG. BR.	2	2	2	2	2	2	2	2	2	2	2	2	
			006290	006291	006292	006293	006294	006295	006296	006297	006298	006299	006300	006301	006302
	2	Grid & Cover Assy. (1)	1	1	1	1	1	1	1	1	1	1	1	1	
			006750	006751	006752	006753	006754	006755	006756	006757	006758	006759	006760	006761	006762
Type T20	1	Std. Hub – RG. BR.	2	2	2	2	2	2	2	2	2	2			
			006290	006291	006292	006293	006294	006295	006296	006297	006298	006299			
	3	Grid & Cover Assy. (1)	1	1	1	1	1	1	1	1	1	1			
			006765	006766	006767	006768	006769	006770	006771	006772	006773	006774			
Type T31	4	Shaft Hub – RG. BR.	2	2	2	2	2	2	2	2	2	2			
			006305	006306	006307	006308	006309	006310	006311	006312	006313	006314			
	2	Grid & Cover Assy. (1)	1	1	1	1	1	1	1	1	1	1			
			006750	006751	006752	006753	006754	006755	006756	006757	006758	006759			
	5	Spacer Hub Assy. (2)	SEE TABLE 7												
Type T35	1	Std. Hub – RG. BR.	1	1	1	1	1	1	1	1	1	1			
			006290	006291	006292	006293	006294	006295	006296	006297	006298	006299			
	2	Grid & Cover Assy. (1)	1	1	1	1	1	1	1	1	1	1			
			006750	006751	006752	006753	006754	006755	006756	006757	006758	006759			
	4	Shaft Hub – RG. BR.	1	1	1	1	1	1	1	1	1	1			
			006305	006306	006307	006308	006309	006310	006311	006312	006313	006314			
	5	Spacer Hub Assy. (2)	SEE TABLE 6												

(1) Includes Grid, Cover Halves, Fasteners, Seals and Gasket

(2) Includes Spacer Hub and Fasteners


Table 5 – Recommended Spare Parts for GRID-LIGN Couplings

Cplg. Type	Name of Part	Part Number and Quantity Required per Coupling Size (1)									
		1020T	1030T	1040T	1050T	1060T	1070T	1080T	1090T	1100T	1110T
T10 T31 T35	Cover Accessory Kit (2)	1	1	1	1	1	1	1	1	1	1
		006715	006715	006715	006716	006716	006716	006717	006717	006718	006718
	Spacer Hub Accessory Kit	(5)	(5)	(5)	(5)	(5)	(5)	(5)	(5)	(5)	(5)
		006727	006728	006728	006729	006730	006731	006732	006733	006734	006734
	Grid	1	1	1	1	1	1	1	1	1	1
		006275	006276	006277	006278	006279	006280	006281	006282	006283	006284
	Cover Assembly (3)	1	1	1	1	1	1	1	1	1	1
T20		006250	006251	006252	006253	006254	006255	006256	006257	006258	006259
	Cover Seal	2	2	2	2	2	2	2	2	2	2
		006630	006631	006632	006633	006634	006635	006636	006637	006638	006639
	Cover Gasket	2	2	2	2	2	2	2	2	2	2
		006690	006691	006692	006693	006694	006695	006696	006697	006698	006699
	Cover Accessory Kit	1	1	1	1	1	1	1	1	1	1
		006721	006722	006722	006723	006723	006723	006724	006724	006725	006725
T20	Grid	1	1	1	1	1	1	1	1	1	1
		006275	006276	006277	006278	006279	006280	006281	006282	006283	006284
	Cover Assembly (4)	1	1	1	1	1	1	1	1	1	1
		006260	006261	006262	006263	006264	006265	006266	006267	006268	006269
	Cover Seal	2	2	2	2	2	2	2	2	2	2
		006645	006646	006647	006648	006649	006650	006651	006652	006653	006654
	Cover Gasket	1	1	1	1	1	1	1	1	1	1
		006700	006701	006702	006703	006704	006705	006706	006707	006708	006709

(1) For 1120T-1140T, consult DODGE/Reliance Electric.

(2) Includes Fasteners only

(3) Includes Cover Halves, Fasteners, Seals and Gaskets

(4) Includes Cover Halves, Fasteners, Seals and Gasket

(5) 2 Required for T31; 1 Required for T35

Table 6 — Stock GRID-LIGN T35 Half Spacer Coupling Spacer Hub Assemblies

B.S.E. Dimension (in.)	Coupling Size (1)												1080T		1090T		1100T	
	1020T		1030T		1040T		1050T		1060T		1070T							
	Spacer Hub Assembly		Spacer Hub Assembly		Spacer Hub Assembly		Spacer Hub Assembly		Spacer Hub Assembly		Spacer Hub Assembly		Spacer Hub Assembly		Spacer Hub Assembly			
	Part #	Qty.	Part #	Qty.	Part #	Qty.	Part #	Qty.	Part #	Qty.	Part #	Qty.	Part #	Qty.	Part #	Qty.		
1.78	006497	1	006504	1	006516	1												
2.22	006498	1	006505	1	006517	1	006533	1										
2.53	006499	1	006506	1	006518	1	006534	1	006544	1	006533	1						
3.50					006519	1	006535	1										
3.66			006507	1	006520	1	006536	1	006545	1	006554	1	006561	1	006569	1		
4.06																		
4.94																006574	1	
5.00																006575	1	
6.16									006546	1	006555	1	006563	1	006570	1		

(1) For 110T-1140T, consult DODGE/Reliance Electric for delivery information.

Table 7 — Stock GRID-LIGN T31 Full Spacer Coupling Spacer Hub Assemblies

B.S.E. Dimension (in.)	Coupling Size (1)																			
	1020T		1030T		1040T		1050T		1060T		1070T		1080T		1090T		1100T			
	Spacer Hub Assembly		Spacer Hub Assembly		Spacer Hub Assembly		Spacer Hub Assembly		Spacer Hub Assembly		Spacer Hub Assembly		Spacer Hub Assembly		Spacer Hub Assembly		Spacer Hub Assembly			
	Part #	Qty.	Part #	Qty.	Part #	Qty.	Part #	Qty.	Part #	Qty.	Part #	Qty.	Part #	Qty.	Part #	Qty.	Part #	Qty.		
3.50	006497	2	006504	2	006516	2														
3.94	006497	1	006504	1	006516	1														
	006498	1	006505	1	006517	1														
4.25	006497	1	006504	1	006516	1														
	006499	1	006506	1	006518	1														
4.38	006498	2	006505	2	006517	2													006533	2
4.69	006498	1	006505	1	006517	1													006533	1
	006499	1	006506	1	006518	1													006534	1
5.00	006499	2	006506	2	006518	2	006534	2	006544	2	006553	2								
5.22					006516	1														
					006519	1														
5.38			006504	1	006516	1														
			006507	1	006520	1														
5.66					006517	1						006533							1	
					006519	1						006535							1	
5.81			006505	1	006517	1						006533							1	
			006507	1	006520	1						006536							1	
5.97					006518	1	006534	1												
					006519	1	006535	1												
6.12			006506	1	006518	1	006534	1	006544	1	006553	1								
			006507	1	006520	1	006536	1	006545	1	006554	1								
6.94					006519	2	006535	2												
7.09					006519	1	006535	1												
					006520	1	006536	1												
7.25			006507	2	006520	2	006536	2	006545	2	006554	2	006561	2	006569	2				
8.00																			006574	2
8.59													006561	1						
													006562	1						
8.62									006544	1	006553	1								
									006546	1	006555	1								
8.88																				
7.25									006545	1	006554	1	006561	1	006569	1	006574	1		
									006546	1	006555	1	006563	1	006570	1	006575	1		
9.94													006562	2						
11.09													006562	1						
													006563	1						
12.25									006546	2	006555	2	006563	2	006570	2				

(1) For 1110T-1140T, consult DODGE/Reliance Electric for delivery information.

Table 8 — Set Screw Installation Torque for Straight and Finished Bore Flanges

Important — Verify Set Screw Size Prior to Tightening

Set Screw Size	#0	#1	#2	#3	#4	#5	#6	#8	#10	1/4	5/16	3/8	7/16	1/2	9/16	5/8	3/4	7/8	1
Installation Torque (in.-lbs.)	1.0	1.8	1.8	5	5	10	10	20	36	87	165	290	430	620	620	1,325	2,400	5,200	7,200



**OPERATION
&
MAINTENANCE
MANUAL
FOR
THREE PHASE
INDUCTION
MOTORS**

TECO-Westinghouse Motor Company
5100 North IH-35
Round Rock, Tx. 78681

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1. INTRODUCTION

This and the following instruction address the more common situations encountered in motor installation, operation and maintenance. For the TWMC motor warranty to be and to remain in effect, the motor must be installed and operated in strict accordance with the outline drawing, motor nameplates and these instructions and must not be altered or modified in any unauthorized manner.

During the installation and operation of motors in heavy industrial applications there is a danger of live electrical parts and rotating parts. Therefore to prevent injury and/or damage the basic planning work for installation, transportation, assembly, operation, etc... needs to be done and checked by authorized and competent personnel only.

Since these instructions cannot cover every eventuality of installation, operation and maintenance, the following points should be considered and checked.

- The technical data and information on permissible use such as assembly, connection, ambient and operating conditions given in the related catalogue, operating instructions, nameplates and other production documentation.
- The general erection and safety regulations.
- The local and plant-specific specifications and requirements.
- The proper use of transport, lifting devices and tools.
- The use of personal protective equipment.

Following indications should be observed when reading these instructions.

Safety instructions are marked as follows:



Warning of electric hazards for personnel.



Warning of dangers for personnel.

ATTENTION! Warning of damage for the motor or installation.
--

2. ACCEPTING, INSPECTION, STORAGE, TRANSPORTATION

Inspection upon receipt

Check to following points upon receipt:

- Are the nameplate ratings identical with what you ordered?
- Are dimensions and color in compliance with your specifications?
- Are the nameplate ratings for space heater, thermal protector, temperature detector, etc. identical with what you ordered?
- Is there any damage?
- Are all accessories and accompanying instruction manuals in good order?
- Please ensure that the arrow head indicator really indicates direction of rotation.
- If there are any specific requirements, please ensure they are in conformity with your specifications.

2.1 Storage

When motors are not in operation, the following precautionary measures should be undertaken to assure best performance.

2.2 Place

- (a) High and dry, well ventilated without direct sun, dust or corrosive gas.
- (b) Not located near to a boiler or freezer.
- (c) Entirely free from vibration and easy for movements.
- (d) Motors should be put on pallets to prevent moisture.

2.3 Moisture prevention

Since moisture can be very detrimental to electrical components, the motor temperature should be maintained about 3°C above the dew point temperature by providing either external or internal heat. If the motor is equipped with space heaters, they should be energized at the voltage shown by the space heater nameplate attached to the motor. Incandescent light bulbs can be placed within the motor to provide heat. However, if used, they must not be allowed to come in contact with any parts of the motor because of the concentrated hot spot that could result.

2.4

Even during storage, the insulation resistance should be kept above the specified values.

- (a) For measurement of insulation resistance and acceptable standard values, please refer to measures stated in 3.1.2 "Measurement of insulation resistance".
- (b) Insulation resistance test should be performed once every three months.

2.5

If the motor is not in operation for a long period (one week and above) after installation or has been in operation but stopped for a period of time, the following precautions must be taken.

- (a) Protect the motor as measures stated in 2.2.3.
- (b) Insulation resistance test should be performed as stated in 2.2.4.

2.6 Bearing protection

- (a) If the motor has been provided with a shaft shipping brace to prevent shaft movement during transit, it must be removed before operating the motor. It is very important that this brace be re-installed exactly as it was originally, before the motor is moved from storage or any time when the motor is being transported. This prevents axial rotor movement that might damage the bearings.
- (b) Motors equipped with sleeve bearings are shipped from the factory with the bearing oil reservoirs drained. In storage, the oil reservoirs should be properly filled to the center of the oil level gauge with a good grade of rust inhibiting oil. To keep the bearing journals well oiled and to prevent rusting, the motor shaft should be rotated several revolutions about every month ensuring the shaft does not come to rest in its original position. While the shaft is rotating, it should be pushed to both extremes of the endplay.
- (c) Motors with anti-friction bearings are properly lubricated with the correct grade of grease at the factory and no further greasing is required in storage. The shaft should be rotated several revolutions about every month to maintain proper distribution of the grease within the bearings.
- (d) Tilt-pad bearings are a type of sleeve bearing used in special design applications. Due to the nature of this bearing, a loose oil ring for delivering lubricant cannot be provided. Therefore, during the storage interval, oil must be periodically manually introduced into the pads and housing to prevent the occurrence of oxidation of the precision machined components.
 - (1) Remove the pipe plug from the bearing cap located above the tilt-bearing shell.
 - (2) Pour in approximately one cup of oil every month and rotate the shaft a few revolutions about every two (2) weeks.
 - (3) For long periods of storage, the oil that accumulates in the housing should be removed.

ATTENTION!

Care should be taken to keep parts such as fitting surfaces, key, shaft extension and axial central hole from any collision with foreign matter. Grease should also be generously applied to prevent rusting.

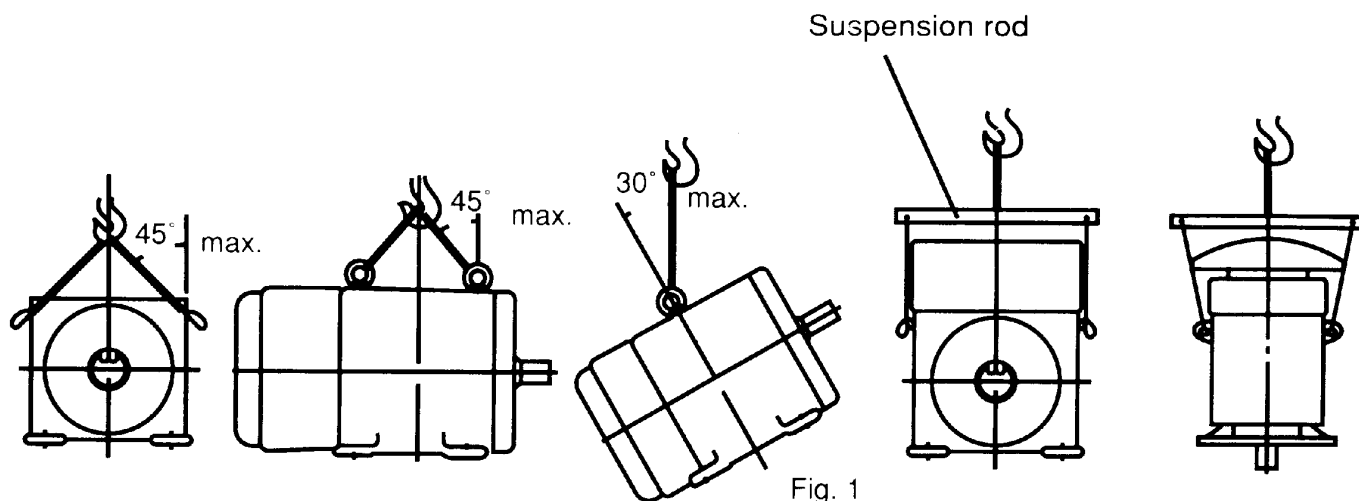
2.7 Transportation

To keep the rotating parts of motors from moving, thus preventing damage and scratching during transportation, they should be held securely with a locking device. Remove all transit clamps before operating the motor. It is very important that this device be reinstalled exactly as it was originally, before the motor is moved from storage or any time when the motor is being transported. The vertical mounting type motors should be transported in the vertical position.



Do not use the hoisting hook/eyebolts to lift more than the motor itself. They are designed to support the motor only. Make sure the hoisting hook is correctly attached to the eyebolt(s)/lug(s) are fully screwed in before hoisting. Also note such parts as fan cover, ventilation box, bracket, slip-ring, etc. may have their own hoisting lugs which can only carry their own weight. Nothing extra should be attached while hoisting.

Do not twist the steel wires and make sure the eyebolts have been firmly screwed and the sling angle is correct.



3 INSTALLATION

Site and environment for motor installation

3.1.1

Standard environment and site conditions for the installation of motors are usually set as follows:

- (a) Ambient temperature: $-10\sim 40^{\circ}\text{C}$
- (b) Humidity: Relative humidity below 90%RH for totally enclosed types, and below 80%RH for semi-enclosed types.
- (c) Elevation: below 1000 meters or 3300 feet.
- (d) Harmful gases, liquids, dusts, high moisture should be absent.
- (e) Foundations should be strong and free of vibration.

If there are any special environmental conditions, please inform TWMC prior to ordering.

3.1.2 Ventilation and space

- (a) Installation area should be well ventilated.
- (b) The installation space should be large enough to facilitate heat dissipation and maintenance.

3.2 Foundation

3.2.1

Use rigid and solid sole plate or common bed as foundation.

For best motor performance, it is advisable to use a sole plate or common bed, particularly when using a shaft coupling.

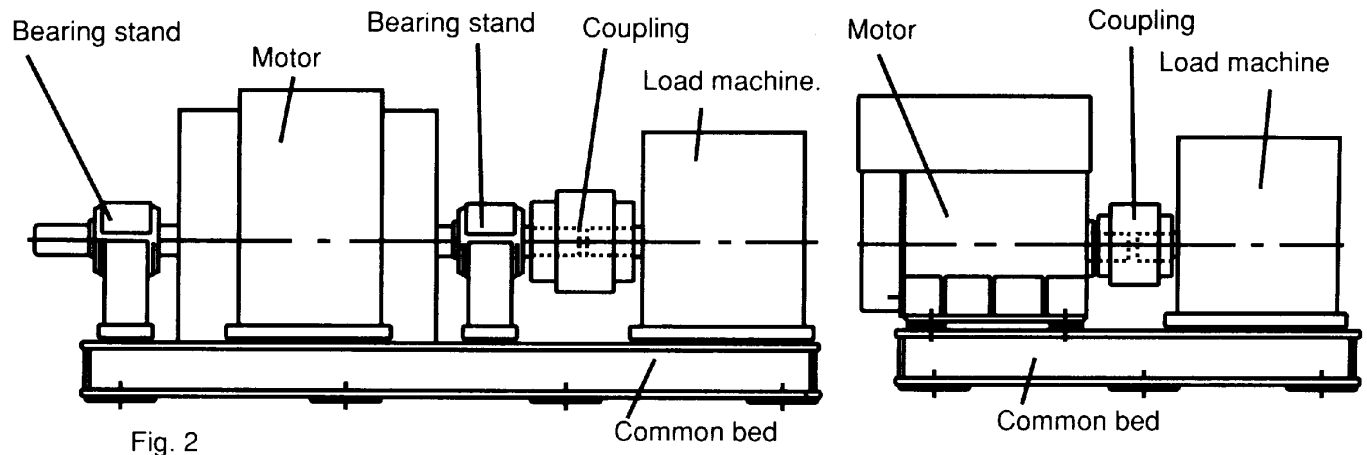


Fig. 2

3.2.2 Installation

- Select an appropriate foundation surface for the sole plate or common bed, which will be, considered the ultimate level.
- Align the position of the common bed with reference to that level.
- Align the level accuracy at least at four points such as bearing mounting, shaft extension etc. The accuracy should be within 0.04mm or .0015 inches
- Sole plate or common bed should be embedded in concrete foundation as illustrated in Fig. 3. Stiff pads should also be installed beneath the wedges, which are welded together at various spots about 400-500mm (15.75-19.70 inches) apart etc., to enable the foundation to carry evenly the weight of the whole motor.
- The base should be sturdy and rigid to keep it flat and level.
- Make sure the mortar and concrete are completely dry, and the precision of the level is acceptable, and then set the motor on the mounting foundation.
- Accurately install shaft couplings, belt sheaves etc., then weld the wedges solid to prevent untoward change in position.

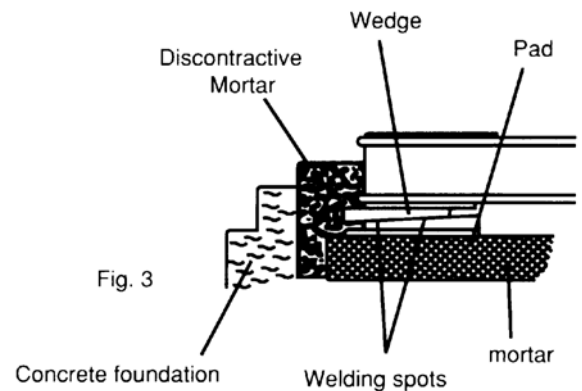


Fig. 3

3.2.3 The foundation of vertical induction motors: (Also the foundation of pump)

- Foundation of motor/pump must be rigid and secure to provide adequate support. There must be no vibration, twisting, misalignment etc. due to inadequate foundations.
- A massive concrete foundation is preferred in order to minimize vibration. Rigidity and stability are enhanced by prop plate and foundation bolt. As shown in Fig. 4.

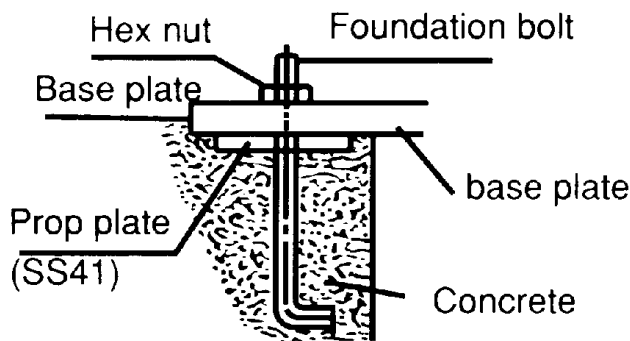


Fig. 4

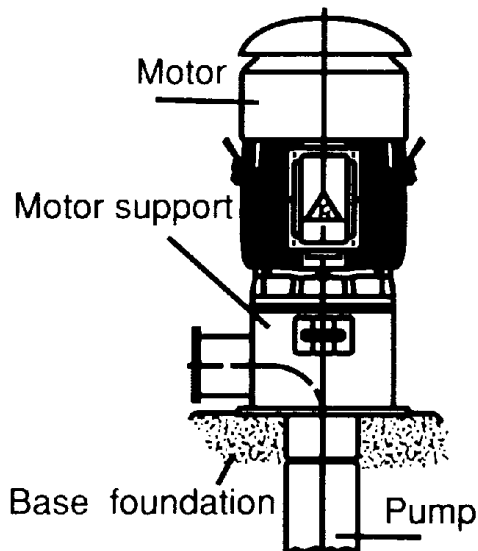


Fig. 5

3.2.4 Installation of vertical motors:

- All mounting surfaces must be clean and level.
- Foundation must be leveled at least at 4 points and guaranteed to be below 0.04mm (.0015 in.) flat and level.
- Make sure the mortar and concrete are completely dry, and the precision of the level is acceptable, and then set the motor on the mounting foundation.
- Accurately install shaft couplings.

3.3 Installation of shaft coupling

ATTENTION!

Motors must always be accurately aligned, and this applies especially where they are directly coupled.

Incorrect alignment can lead to bearing failure, vibration and even shaft fracture. As soon as bearing failure or vibration is detected, the alignment should be checked.

3.3.1

Field application of a coupling to the motor shaft should follow the procedures recommended by the coupling manufacturer. The motor shaft extension must not be subjected to either extreme heat or cold during coupling installation.

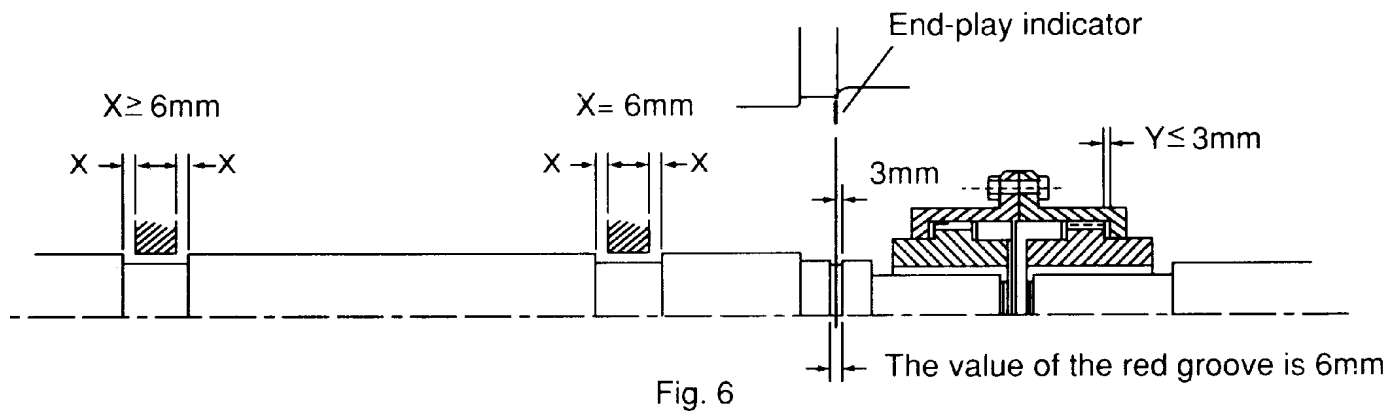
ATTENTION!

Basically, the coupling should be heated and pushed onto the shaft extension with slight axial force. Do not hammer coupling to prevent bearing damage.

3.3.2

Although the sleeve bearings are equipped with thrust faces, these are intended only to provide momentary axial restraint of rotor movement either during start-up or when operating the motor disconnected from the driven equipment. They must not be operated under a constant thrust load unless they were originally designed for this condition.

Motors with either sleeve or anti-friction bearings are suitable for connection to the driven load through a flexible coupling. Coupling solidly to the load is not acceptable. With sleeve bearings, the flexible coupling should be of the limited end float type to prevent the possibility of any end thrust from load being transmitted to the motor bearings, which could cause bearing damage. The recommended limits of end float are as follows:



- When the motor is in operation after installation, be sure that the end-play indicator is within the 6mm (.236 in.) of the groove on the shaft or aligned to the shaft shoulder immediately outboard of the drive-end bearing to assure there is low friction between shaft and bearing.
- Unless otherwise specified, the designed end-play value X of the groove for TWMC motors in general is within 6mm (.236 in.) as illustrated in Fig. 6. In essence, the endplay indicator is adjusted to point at the center of the groove or the drive-end shaft shoulder; thus X equals to $6 \pm 1\text{mm}$ or so, and the endplay value (Y) of the couplings should equal or be smaller than 3mm (.118 in.).
- If the desired value Y is greater than 3mm (.118 in.) caused for instance by a thrust load and/or load machine with large end-play, please inform TWMC prior to entering an order.

3.3.3

In aligning the motor (and rotor) axially with the driven equipment, consideration should be given not only to the endplay indicator position but also to axial shaft expansion and increase in shaft centerline height due to thermal effects. In general, the axial shaft growth for motors can be disregarded since neither bearing is fixed and any shaft growth due to temperature increase will produce an elongation away from the coupling.

Shaft height growth (change in shaft centerline elevation) for TEFC machines can be calculated as follows:

$$\Delta = (0.0005") \times (\text{motor foot to shaft } \text{£} \text{ dimension})$$

For non-TEFC machines, divide the number by 2.

3.3.4

It is desirable, in normal operation that the motor operates on its magnetic center, so that no axial force is exerted on the coupling.

The motor shaft and the driven shaft should be aligned within the following tolerances in both angular and parallel alignment:

Unit: mm

TIR	Range of rotating speed	Solid coupling	Flexible coupling
C	2500 rpm and above	0.03	0.03
	Below 2500 rpm	0.04	0.05
A	2500 rpm and above	0.03	0.03
	Below 2500 rpm	0.03	0.04

Angular misalignment is the amount by which the centerlines of driver and driven shafts are skewed. It can be measured using a dial indicator set up as shown in Fig. 7. The couplings are rotated together through 360 degrees so that the indicator does not measure runout of the coupling hub face. The shafts should be forced against either the in or out extreme of their end float while being rotated.

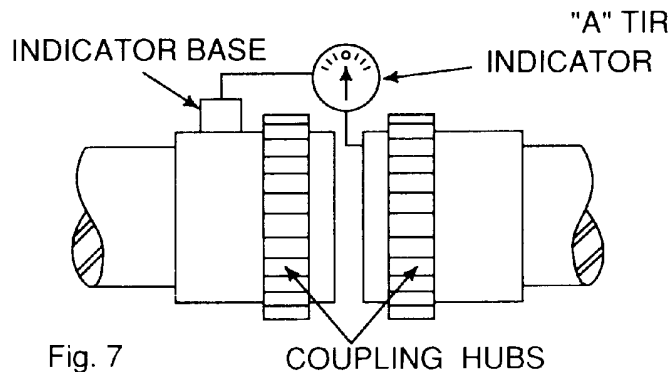


Fig. 7
TIR=Total indicator reading (by dial indicator)

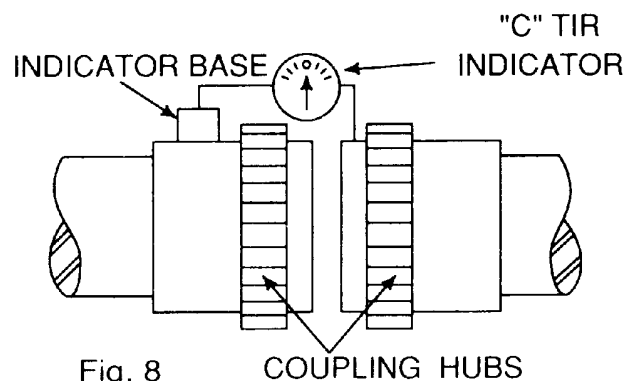


Fig. 8

Parallel misalignment is the amount by which the centerlines of the driver and driven shafts are out of parallel. It can be measured using a dial indicator set up as shown in Fig. 8. Again, the couplings are rotated together through 360 degrees so that the indicator does not measure runout of the coupling hub outside diameter.

3.3.5

After the motor has been properly aligned with the driven equipment and the hold-down bolts have been installed and tightened, for motors with fabricated frames, at least two dowel pins should be installed in two diagonally opposite motor feet.

3.3.6 Installation of shaft coupling: (Vertical hollow shaft motor only)

Bolted Coupling as shown in Fig. 9

- Bearings are provided to absorb some upward shaft thrust when the coupling is fitted.
- The coupling is fastened with bolts.
- This coupling type is not auto-release type.

Note: Standard high thrust motors can absorb momentary up-thrust load up to 30% of the standard down thrust load. If the up-thrust is long in duration (over 10 Seconds) and/or exceeds 30% of the standard high thrust rating, special design arrangements are required and standard motor is not suitable.

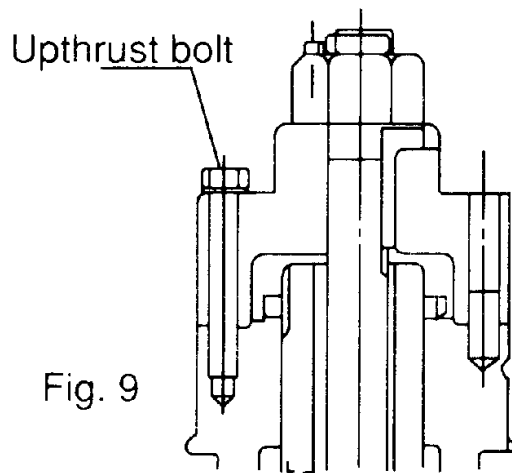


Fig. 9

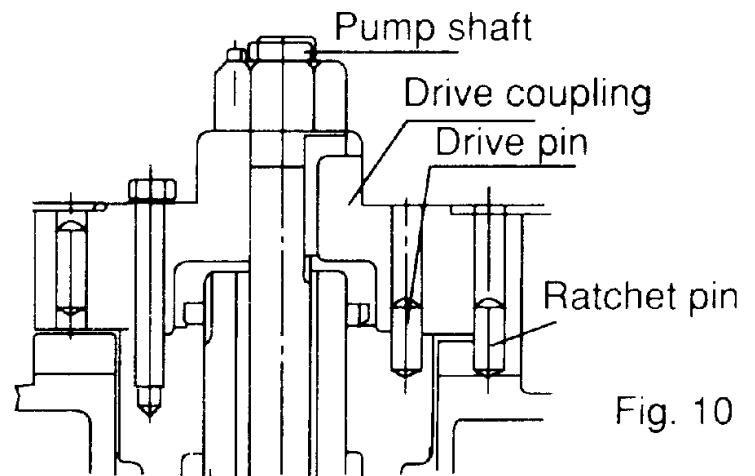


Fig. 10

3.3.7 Non-reverse ratchet/coupling, as Fig. 10 (If necessary)

The non-reverse coupling is also a bolted type and,

- (a) It prevents the pump and motor from rotating in the reverse direction.
- (b) It also prevents damage form over speeding and damage to pump shaft and bearings.
- (c) The ratchet pins are lifted by the ratchet teeth and are held clear by centrifugal force and friction as the motor comes up to speed.
- (d) When power is removed, speed decreases, and the pins fall. At the instant of reversal, a pin will catch in a ratchet tooth and prevent backward rotation.
- (e) When installing the non-reverse coupling, do not use lubricant. Lubricant will interfere with proper operation. The top half of the coupling should seat solidly on the lower half and the pins should touch the bottom of the pockets between the teeth in the plate.
- (f) As with the bolted coupling, the up-thrust capabilities are 30% of the standard high thrust rating for down thrust.

ATTENTION!

Do not apply non-reverse ratchets on applications in which the pump reversal time from shutdown (the instant the stop button is pressed) to zero speed is less than one second.

3.4 Installation for belt drive

In general, power transmission though direct flexible coupling is appropriate for large motors. Such motors are not suitable for belt, chain or gear connection unless specially designed for such service. However, for small and medium motors of which outputs within the ranges shown on table below, it is acceptable to use belt transmission as indicated. Beyond these ranges, do not apply belt sheaves unless specially designed.

3.4.1

The diameter ratio between conveyance sheaves should not be greater than 5 to 1 for flat belts, and 8 to 1 for V-belts. It is also advisable to limit the belt velocity to under 35m/sec (115 ft/sec) to limit belt abrasion and vibration. The smaller the outer diameter of the V-belt sheave, the greater the shaft bending stress will be. If the bending stress is in excess of the shaft fatigue stress, the shaft may break. Therefore, please inform TWMC when you have decided the size of the sheaves and the length of the belts upon ordering.

ATTENTION!

Place the sheave and belt as close as possible to the motor body (it is advisable to make x as shown in Fig. 11 equal to 0) to reduce the bending moment and improve shaft life.

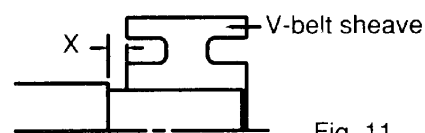


Fig. 11

3.4.2 Table of belt-sheave application for general electric motors

Output (KW/HP)			V-Belt Sheave							
			Conventional V-Belts				Narrow V-Belts			
4P	6P	8P	V-Belt Type	Number Of Belts	Min. PCD (mm)	Max Width (mm)	V-Belt Type	Number Of Belts	Min. PCD (mm)	Max Width (mm)
11/15	-	-	B	4	160	82	3V	4	125	48
-	11/15	-	B	5	170	101	3V	5	140	59
-	-	11/15	B	5	190	101	3V	6	160	69
15/20	-	-	B	5	170	101	3V	6	125	69
-	15/20	-	B	5	224	101	3V	6	160	69
-	-	15/20	C	4	224	111	5V	3	180	60
18.5/25	-	-	B	5	200	101	3V	6	140	69
-	18.5/25	-	C	4	224	111	5V	3	180	60
-	-	18.5/25	C	5	224	136	5V	4	180	78
22/30	-	-	B	5	224	101	5V	6	160	69
-	22/30	-	C	5	224	136	3V	4	180	78
-	-	22/30	C	5	250	136	5V	4	200	78
30/40	-	-	C	5	224	136	5V	4	180	78
-	30/40	-	C	5	265	136	5V	4	224	78
-	-	30/40	C	6	265	162	5V	5	224	95
37/50	-	-	C	6	224	162	5V	4	200	78
-	37/50	-	C	6	265	162	5V	4	224	78
-	-	37/50	C	7	280	187	5V	5	250	95
45/60	-	-	C	6	265	162	5V	4	224	78
-	45/60	-	C	7	280	187	5V	5	224	95
-	-	45/60	C	7	315	187	5V	6	250	113
55/75	-	-	C	7	265	187	5V	5	224	95
-	55/75	-	C	8	300	213	5V	6	250	113
-	-	55/75	D	5	355	196	5V	6	280	113
75/100	-	-	C	8	315	213	5V	6	250	113
-	75/100	-	D	6	355	233	5V	6	315	113
-	-	75/100	D	6	400	233	5V	6	355	113
-	90/120	-	D	6	400	233	5V	6	355	113
-	-	90/120	D	6	425	233	8V	4	355	124
-	110/150	-	D	7	400	270	8V	4	355	124
-	132/175	110/150	D	7	450	270	8V	4	400	124
-	160/200	132/175	D	9	450	344	8V	4	450	124

3.5 Conveyance with chain or gear**3.5.1**

Make sure the loading capacity of shaft and bearings is appropriate for the size and installation position (overhung) of chain and gear. If necessary, please contact us to ensure the shaft and bearings will meet your requirements.

3.5.2

Pay close attention to ensure the parallelism of shafts.

3.5.3

The teeth of couplings should be correctly and precisely matched; the force conveyance centers should lie on the same line.

3.5.4

There should be no skip, jumping, vibration or unusual noises.

ATTENTION!

Do not hammer the conveyance devices such as couplings, belt sheaves, chain wheels, gears etc. onto the shaft. Those shaft fitments should be fitted and removed only by means of suitable devices. Heat shrinking may be a better alternative to avoid damaging bearings and components.



The exposed rotating parts should be covered to prevent accidents.

3.6 Electrical connections

All interconnecting wiring for controls and grounding should be in strict accordance with local requirements such as the USA National Electrical Code and UK IEE wiring regulations. Wiring of motor and control, overload protection and grounding should follow the instructions of connection diagrams attached to the motor.

3.6.1 Power

The rated conditions of operation for the motor are as shown on the nameplate. Within the limits, given below, of voltage and frequency variation from the nameplate values, the motor will continue to operate but with performance characteristics that may differ from those at rated conditions:

±10% of rated voltage

±5% of rated frequency

±10% combined voltage and frequency variation so long as frequency variation is no more than ±5% of rated.

Operating the motor at voltages and frequencies outside of the above limits can result in both unsatisfactory motor performance and damage to or failure of the motor.

3.6.2

The main lead box furnished with the motor has been sized to provide adequate space for the make-up of the connections between the motor lead cables and the incoming power cables.



The bolted joints between the motor lead and the power cables must be made and insulated in a workman-like manner following the best trade practices.

3.6.3

Either fabricated motors or fan cooled cast frame, motors are all provided with grounding pads or bolts.



The motor must be grounded by proper connection to the electrical system ground.

3.6.4

The rotation direction of the motor will be as shown by either a nameplate on the motor or the outline drawing. The required phase rotation of the incoming power for this motor rotation may also be stated. If either is unknown, the correct sequence can be determined in the following manner: While the motor is uncoupled from the load, start the motor and observe the direction of rotation. Allow the motor to achieve full speed before disconnecting it from the power source. Refer to the operation section of these instructions for information concerning initial start-up. If resulting rotation is incorrect, it can be reversed by interchanging any two (2) incoming cables.

3.6.5 Auxiliary devices

Auxiliary devices such as resistance temperature detectors, thermocouples, thermoguards, etc., will generally terminate on terminal blocks located in the auxiliary terminal box on the motor. Other devices may terminate in their own enclosures elsewhere on the motor. Such information can be obtained by referring to the outline drawing. Information regarding terminal designation and the connection of auxiliary devices can be obtained from auxiliary drawings or attached nameplates.

If the motor is provided with internal space heaters, the incoming voltage supplied to them must be exactly as shown by either a nameplate on the motor or the outline drawing for proper heater operation.



Caution must be exercised anytime contact is made with the incoming space heater circuit as space heater voltage is often automatically applied when the motor is shutdown.

4. OPERATION

4.1 Examination before start

4.1.1

When motors are installed in good manner, ensure the wiring is according to the diagram. Also, the following points should be noted:

- (a) Make sure all wiring is correct.
- (b) Ensure the sizes of cable wires are appropriate and all connections are well made for the currents they will carry.
- (c) Ensure all connections are properly insulated for the voltage and temperature they will experience.
- (d) Ensure the capacity of fuses, switches, magnetic switches and thermo relays etc. are appropriate and the contactors are in good condition.
- (e) Make sure the frame and terminal box are grounded.
- (f) Make sure that the starting method is correct.
- (g) Make sure switches and starters are set at their right positions.
- (h) Motor heaters must be switched off when the motor is running.

4.1.2 Measurement of insulation resistance



During and immediately after measuring, the terminals must not be touched as they may carry residual dangerous voltages. Furthermore, if power cables are connected, make sure that the power supplies are clearly disconnected and there are no moving parts.

- (a) For rated voltage below 1000V, measured with a 500VDC megger.
- (b) For rated voltage above 1000V, measured with a 1000VDC megger.
- (c) In accordance with IEEE 43, clause 9.3, the following formula should be applied:

$$R \geq \left(\frac{\text{Rated voltage (v)}}{1000} + 1 \right) \times 10(\text{M}\Omega)$$

- (d) On a new winding, where the contaminant causing low insulation resistance is generally moisture, drying the winding through the proper application of heat will normally increase the insulation resistance to an acceptable level. The following are several accepted methods for applying heat to the winding:
 - (1) If the motor is equipped with space heaters, they can be energized to heat the winding.
 - (2) Direct current (as from a welder) can be passed through the winding. The total current should not exceed approximately 50% of rated full load current. If the motor has only three leads, two must be connected together to form one circuit through the winding. In this case, one phase will carry the fully applied current and each of the others, one-half each. If the motor has six leads (3 mains and 3 neutrals), the three phases should be connected into one series circuit.



Ensure there is adequate guarding so live parts cannot be touched.

- (3) Heated air can either blown directly into the motor or into a temporary enclosure surrounding the motor. The source of heated air should preferably be electrical as opposed to fueled (such as kerosene) where a malfunction of the fuel burner could result in carbon entering the motor.

ATTENTION!

Caution must be exercised, when heating the motor with any source of heat other than self contained space heaters, to raise the winding temperature at a gradual rate to allow any entrapped moisture to vaporize and escape without rupturing the insulation. The entire heating cycle should extend over 15-20 hours.

Insulation resistance measurements can be made while the winding is being heated. However, they must be corrected to 40°C for evaluation since the actual insulation resistance will decrease with increasing temperature. As an approximation for a new winding, the insulation resistance will approximately halve for each 10°C increase in insulation temperature above the dew point temperature.

- (e) Should the resistance fail to attain the specified value even after drying, careful examination should be undertaken to eliminate all other possible causes, if any.

4.1.3 Power Source

- (a) Ensure the capacity of the power source is sufficient.
- (b) Ensure the supply voltage and frequency ratings are identical to those on the nameplate.
- (c) Voltage variation should be confined to within $\pm 10\%$ of the rated value and the phase to phase voltages should be balanced.

4.1.4 Bearing lubrication

- (a) For sleeve bearing motors, the oil reservoir must be filled with oil to the correct level. On self-lubricated bearings, the standstill oil level will be at the center of the oil gauge. The proper oil is a rust and oxidation inhibited, turbine grade oil. Refer to the lubrication nameplate for the recommended viscosity.
- (b) Motors, which are supplied with provision for flood lubrication, have an inlet orifice to meter the oil flow to the bearing. Refer to the outline drawing for these values. If the supply pressure does not match that stated on the outline, the orifice size must be adjusted to produce the specified flow rate. The drain adapter (also provided) has a weir plate fixed to the inside of the pipe to permit the establishment of the proper oil level. This weir plate must be located at the bottom of the pipe and must be parallel to the plane of the motor feet. To ensure optimum flow, the drain line should be vented to the atmosphere.
 - Oil inlet temperature: Normal below 50°C
 - Alarm 60°C
 - Trip 65°C
- (c) If the motor is in storage for over three (3) months, refilling of some new oil should be undertaken before operation to prevent bearing damage due to dry friction. The oil level should be kept at the center of the oil gauge. If necessary, drain some oil after refilling.
- (d) Motors that have been designed with anti-friction bearings for use with an oil mist lubrication system have been packed at the factory with a small amount of grease for short test runs. Continuous running should not be considered unless the oil mist system is installed and operating.
- (e) Grease lubricant type
 - (1) The bearings have been well greased at the factory before delivery. However, regreasing is required if a significant period has elapsed between manufacture and use or in storage
 - (2) **All motors with ZZ bearings will have SHELL Alvania R3 (Lithium base grease).** All motors with open bearings will have Polyrex EM (polyurea base grease).

4.1.5 Cooling water for the cooler on water-cooled motors

Make sure the quality, volume and inlet temperature of cooling water for the motors are normal before the machine is in operation.

Water: General tower water or industrial water.

Volume: Please see outline drawing

Inlet temperature: Normal below 30°C
 Alarm 35°C
 Trip 40°C

ATTENTION!

Make sure all locks, which fasten the movable parts of the motors during transportation, are dismantled and the shaft can rotate freely.

ATTENTION!

Ensure there is no foreign matter or tools inside the motors before starting motors.

4.1.6

Make sure the transmission system, including belts, screws, bolts, nuts and set pins are in good condition.



The keys fitted to the shaft extensions are held by plastic tape only to prevent them from falling out during transportation or handling. The shaft key shall be removed to avoid flying out, when the motor is operated prior to the couplings etc. being fitted to the shaft extension.

4.1.7

Make sure the items above are examined. Test the motor running with or without load. Record and check according to "Maintenance" at 15-minute intervals during the first three hours of operation. Then regular examinations should take place at longer intervals. If all goes well the motor can be classified as "in good order".

4.2 Starting operation

4.2.1 Starting load

Initially run the motor unloaded prior to coupling to other machines. Unless otherwise specified, a motor usually starts with light load, which is then gradually increased, proportional to the square of the speed and at last reaches 100% load at full load speed.

4.2.2 Starting

Too frequent starts can be harmful to the motors. The following restrictions should be observed:

- (a) Motor can be restarted should the initial start fail. Two starts are generally permissible when the motor is cold.
- (b) Motor can be started only once when it is at normal running temperature.
- (c) Should additional starts be necessary beyond the conditions stated above, the following restrictions should be noted:
 - (1) Let the motor cool down for 60 minutes before restarting, fully loaded.
 - (2) Let the motor cool down for 30 minutes before restarting, unloaded.
 - (3) Two inching starts can be regarded as one normal start.

ATTENTION!

If the motor rotor fails to start turning within one or two seconds, shut off the power supply immediately.

Investigate thoroughly and take corrective action before attempting a restart.

Possible reasons for not starting are:

- (1) Too low a voltage at the motor terminals.
- (2) The load is too much for the rotor to accelerate.
- (3) The load is frozen up mechanically.
- (4) All electrical connections have not been made.
- (5) Single-phase power has been applied.
- (6) Any combination of the above.

4.2.3 Rotating direction

- (a) Most TWMC motors are bi-directional. However, when some special types, such as high speed 2-Pole, certain large capacity motors, those with a non-reversing ratchet etc., should rotate in one direction, please ensure the rotation is in conformity with the directional arrow-mark shown on the attached nameplate.
- (b) To reverse a bi-directional motor, cut the power and wait until the motor stops. Then interchange any two of the three phases.

4.2.4 Power source, Voltage, Current

- (a) Ensure the voltage and frequency of the power source are identical to the ratings shown on the nameplate.
- (b) Voltage variation should be confined to within $\pm 10\%$ of the rating and the three phase voltages should be in full balance
- (c) Ensure the motor phase currents, when without load, are within $\pm 5\%$ of the average values.

4.2.5

Frequency variation should be confined to within $\pm 5\%$ of the rating. The aggregate variation of voltage and frequency should be confined to within $\pm 10\%$ of the absolute value of the ratings.

Starting time and unusual noises

ATTENTION!

Starting time is longer for the motors with large inertia. However, if starting time is longer than usual or if there is difficulty in starting, or there is abnormal noise, do not run the motor and refer to TWMC Service representative.

4.2.6 Sleeve bearing oil rings (sleeve bearing types only)

As the oil ring is used to carry lubricant to sleeve bearings, frequently check to ensure the oil ring is in motion.

4.2.7 Bearing temperature rise

Following the initial start-up, the bearing temperatures should be closely monitored. The rate of rise in bearing temperature is more indicative of impending trouble than is the actual temperature.

ATTENTION!

If the rate of rise in temperature is excessive or if the motor exhibits excessive vibration or noise, it should be shut down immediately and a thorough investigation made as to the cause before it is operated again.

If the bearing temperature rise and motor operation appear to be normal, operation should continue until the bearing temperature stabilizes.

Recommended limits on bearing temperature are as follows:

Sleeve Bearings

- By permanently installed detector
- By temporary detector on top of the bearing sleeve near the oil ring

Total measured temperature

90°C
85°C

Anti-Friction Bearings

- By permanently installed detector
- By temporary detector measuring the outside of the bearing housing

Total measured temperature

100°C
95°C

ATTENTION! (For sleeve bearing)

- (1) It must be noted that when operating flood lubricated sleeve bearings without outside lubrication supplied, the bearing temperature must not be allowed to exceed 85°C total temperature
- (2) Under normal condition, for the self-lube bearing, the rate of temperature rise should be from 11 to 14°C for the first ten (10) minutes after starting up and approximately 22°C at thirty (30) minutes. The rate of bearing temperature rise is a function of the natural ventilation and operating conditions.
- (3) When the rate of bearing temperature rise is less than 1°C per half-hour, the bearing temperature is considered to be stabilized.
- (4) If the total bearing temperature exceeds 95°C, the motor should be shut down immediately.

Noise and Vibration

ATTENTION!

Any abnormal noise or vibration should be immediately investigated and corrected. Increased vibration can be indicative of a change in balance due to mechanical failure of a rotor part, a stator winding problem or a change in motor alignment.

5. MAINTENANCE

5.1 Major points in regular inspections and maintenance.



For safety, maintenance and repairs must only be carried out by properly trained personnel.



Some testing, such as insulation resistance, usually requires the motor to be stopped and isolated from power supply(ies).

Routine inspection and maintenance are usually performed by looking, listening, smelling and simple meters.



High temperature may arise under operating conditions on the motor surfaces, so that touching should be prevented or avoided. Keep away from moving and live parts. Unless deemed necessary, do not remove guards whilst assessing the motor.

Timely replacement of worn parts can assure longevity and prevent breakdown.

Routine inspection and regular inspection and maintenance are important in preventing breakdown and lengthening service life.

Owing to the varied time and circumstances, motors are used, it is difficult to set the items and periods for regular inspection and maintenance. However, as a guide it is recommended to be performed periodically according to factory maintenance program. Generally, the inspection scope determined by the following factors:

- (a) Ambient temperature.
- (b) Starting and stopping frequency.
- (c) Troublesome parts usually affecting motor functions.
- (d) Easily abraded parts.
- (e) The important position of motor in the operational system of a factory should be duly recognized. Therefore, its health and wellbeing should be fully protected especially when it is operating in severe conditions.

5.2 Motor windings:

- (a) Measurement of insulation resistance and standards to determine quality of insulation resistance, please refer to measures stated in 3.1.2 "Measurement of insulation resistance".
- (b) Inspection of coil-ends:
 - (1) Grease and dust accumulated on coils may cause insulation deterioration and poor cooling effect.
 - (2) Moisture must not accumulate. Keep coils warm when motor is not in use if moisture can be seen.
 - (3) Discoloring. This is mainly caused by overheating.
- (c) Ensure no untoward change of wedges from original position.
- (d) Ensure the binding at the coil end is in its normal position.

5.3 Clean the interior of the motor:

- (a) After a motor is in operation for some time, accumulation of dust, carbon powder and grease etc., on the inside is unavoidable, and may cause damage. Regular cleaning and examination is necessary to assure top performance.
- (b) Points to note during cleaning:
 - (1) If using compressed air or blower:
 - (a) Compressed air should be free of moisture.
 - (b) Maintain air pressure at 4 kg/cm², since high pressure can cause damage to coils.
 - (2) Vacuum
Vacuum cleaning can be used, both before and after other methods of cleaning, to remove loose dirt and debris. It is a very effective way to remove loose surface contamination from the winding without scattering. Vacuum cleaning tools should be non-metallic to avoid any damage to the winding insulation
 - (3) Wiping
Surface contamination on the winding can be removed by wiping using a soft, lint-free wiping material. If the contamination is oily, the wiping material can be moistened (not dripping wet) with a safety type petroleum solvent. In hazardous locations, a solvent such as inhibited methyl chloroform may be used, but must be used sparingly and immediately removed. While this solvent is non-flammable under ordinary conditions, it is toxic and proper health and safety precautions should be followed while using it.

ATTENTION!

Solvents of any type should never be used on windings provided with abrasion protection. Abrasion protection is a gray, rubber-like coating applied to the winding end-turns.



Adequate ventilation must always be provided in any area where solvents are being used to avoid the danger of fire, explosion or health hazards. In confined areas (such as pits) each operator should be provided with an airline respirator, a hose mask or a self-contained breathing apparatus. Operators should wear goggles, aprons and suitable gloves. Solvents and their vapors should never be exposed to open flames or sparks and should always be stored in approved safety containers.

- (4) Keep core ducts completely clean. The difference in temperature rise could be around 10°C before and after cleaning

5.4 Clean the exterior of the motor:

- (a) On open ventilated motors, screens and louvers over the inlet air openings should not be allowed to accumulate any build-up of dirt, lint, etc. that could restrict free air movement.

ATTENTION!

Screens and louvers should never be cleaned or disturbed while the motor is in operation because any dislodged dirt or debris can be drawn directly into the motor.

- (b) If the motor is equipped with air filters, they should be replaced (disposable type) or cleaned and reconditioned (permanent type) at a frequency that is dictated by conditions. It is better to replace or recondition filters too often than not often enough.
- (c) Totally enclosed air to air cooled and totally enclosed fan cooled motors require special cleaning considerations. The external fan must be cleaned thoroughly since any dirt build-up not removed can lead to unbalance and vibration. All of the tubes of the air-to-air heat exchanger should be cleaned using a suitable tube brush having synthetic fiber bristles (not wire of any type).

5.5 Maintenance of anti-friction bearings**5.5.1 Frequency of re-lubrication:**

The life of grease varies greatly as a result of types of model, revolution speed, temperature, operational conditions etc. It is, therefore, impossible to be precise about replenishment intervals. However, for normal direct coupling transmission, the periods shown as Table 1 may be used as a guide.

Remarks:

- (a) The periods shown in Table 1 should be halved where bearings are used for belt drive and/or in dirty or high ambient temperature or high humidity environments.
- (b) Please refer to the lubrication nameplate, if attached to the motor.
- (c) For bearing numbers outside the range of Table 1, please contact TWMC

- (d) If the periods referred to in Table 1 for drive-end bearing and opposite drive-end are different, for the convenience of maintenance operation, please take the shorter one the required grease replenishment period of these bearings.

5.5.1 Kinds of grease:

All motors with ZZ bearings will have SHELL Alvania R3 (lithium base grease). All motors with open bearings will have Polyrex EM (polyurea base grease).

Certain T-frame models will utilize special grease and will be noted on the lubrication nameplate. Please use identical grease or its equivalents when maintaining lubrication schedule.

ATTENTION!

Do not mix different kinds of grease.

Mixing grease with different type of thickeners may destroy its composition and physical properties. Even if the thickeners are of the same type, possible differences in the additive may cause detrimental effects.

5.5.2 Grease quantity

The amount of grease per replenishment depends on the type, size and construction of the bearings. The maximum amount of one replenishment for each bearing is shown in Table 2.

5.5.3 Re-greasing



If re-lubrication is to be performed when the motor is running, stay clear of rotating parts.

It is advisable to re-grease when the motor is running to allow the new grease to be evenly distributed inside the bearing.

Before re-greasing, the inlet fitting should be thoroughly cleaned to prevent any accumulated dirt from being carried into the bearing with the new grease. The outlet of grease drainage should be opened to allow the proper venting of old grease.

Use a grease gun to pump grease through grease nipple into the bearings. After re-greasing, operate the motor for 10-30 minutes to allow any excess grease to vent out.

TABLE 1.

Bearing Number		600 RPM	720 RPM	750 RPM	900 RPM	1000 RPM	1200 RPM	1500 RPM	1800 RPM	3000 RPM	3600 RPM
62XX 63XX 72XX 73XX	6210										
	12									2000 Hrs.	
	13										
	14									1000 Hrs.	
	15										
	16								720 Hrs.		
	17							2000 Hrs.			
	18	3000 Hrs.									
	20										
	22										
	24							1500 Hrs.			
	26										
	28						2000 Hrs.		1000 Hrs.		
	30										
	32								500 Hrs.		
	34						1500 Hrs.				
38				2000 Hrs.		1000 Hrs.					

Bearing Number		600 RPM	720 RPM	750 RPM	900 RPM	1000 RPM	1200 RPM	1500 RPM	1800 RPM	
NU2XX NU3XX	NU214									
	15								2000 Hrs.	
	16									
	17									
	18	3000 Hrs.							1500 Hrs.	
	20									
	22								1000 Hrs.	
	24									
	26					2000 Hrs.				
	28								500 Hrs.	
	30									
	32									
	34				2000 Hrs.		1000 Hrs.			
	36									
	38	2000 Hrs.								
	40									
	44				1000 Hrs.					
	48	1000 Hrs.								

Bearing Number		600 RPM	720 RPM	750 RPM	900 RPM	1000 RPM	1200 RPM	1500 RPM	1800 RPM
222XX 223XX	22220							300 Hrs.	
	22								
	24	1000 Hrs.						500 Hrs.	
	26								
	28								
	30					300 Hrs.			
	32					500 Hrs.			
	34								
	36								
	38	500 Hrs.							
	40					300 Hrs.			
	44								
	48	300 Hrs.							

TABLE 2.

Bearing No.	Amount of replenishment
62XX 72XX NU2XX 2222XX	6210 30 g
	6212 40
	6213 50
	6214 50
	6215 60
	6216 60
	6217 80
	6218 80
	6220 100
	6222 120
	6224 120
	6226 140
	6228 160
	6230 180
	6232 200
	6234 250
	6236 300
	6238 350
	6240 400
	6244 450
	6248 500

Bearing No.	Amount of replenishment
63XX 73XX NU223XX 223XX	6310 40 g
	6312 60
	6313 80
	6314 80
	6315 100
	6316 100
	6317 120
	6318 120
	6320 160
	6322 220
	6324 270
	6326 300
	6328 400
	6330 450
	6332 500
	6334 600
	6336 700
	6338 800
	6340 900
	6344 900
	6348 900

*Fill new grease until it overflows and the old grease is entirely replaced.

5.5.4 Oil re-lubrication (For oil lubrication types only)

Maintain proper lubrication by checking the oil level periodically and adding oil when necessary. Because of the initial clearing action of the bearing and the expansion of the oil as it comes up to operating temperature, the oil level will be higher after the motor has been in operation for a while than it is with the motor at standstill.

Overfilling should be avoided not only because of the possibility that expansion may force the oil over the oil sleeve and on to the rotor, but also because too high an operating oil level prevents the bearing from clearing itself of excess oil. The resultant churning can cause extra loss, high temperatures, and oxidized oil. If, during operation, the oil level goes above the maximum shown on the sight gauge, drain enough oil to bring the level back within the recommended operating range. **Do not permit the operating level to fall below the minimum shown on the gauge.**

ATTENTION!

Should it ever become necessary to add excessive amount of make-up oil, investigate immediately for oil leaks.

Change the oil at regular intervals. The time between oil changes depends upon the severity of operating conditions and, hence, must be determined by the motor user. Two or three changes a year is typical, but special conditions, such as high ambient temperature, may require more frequent changes. Avoid operating the motor with oxidized oil.

Use only good grade, oxidation-corrosion-inhibited turbine oils produced by reputable oil companies.

The viscosity of the oil to be used depends upon the type and size of the bearings, its load and speed, the ambient temperature, and the amount and temperature of the cooling water (if used)). The lubrication nameplate or instructions with each motor specifies the viscosity range of oil suitable for average conditions. The usual oil viscosity range of oil suitable for average conditions. The usual oil viscosity recommendations are summarized in Table 3. Operation in ambient temperatures that are near or below freezing may require preheating the oil or the use of special oil. Whenever the motor is disassembled for general cleaning and reconditioning, the bearing housing may be washed out with a suitable cleaning solvent. Be sure that the oil-metering hole is clear, and then dry the housing thoroughly before re-assembly, and ensure all traces of cleaning solvent have been removed.

TABLE 3 Oil Viscosity**

Bearing function and location	Bearing Type	Oil Viscosity - SSU	
		@ 100°F	@ 200°F
Thrust Bearing	72XX, 73XX Angular contact ball And/or (62XX, 63XX)	150	45
	Spherical roller	300	53
	Plate (Kingsbury Type)	300	53

**Remark: When a lubrication nameplate attached to the motor, use lubrication oil it stipulates.

5.5.5 Cleaning and installation of bearings

- (a) Apply the proper amount of grease to the disassembled parts of the bearing after they have been thoroughly cleaned with high quality cleaning oil. Then protect them from contamination before and during assembly.
- (b) Bearing installation

ATTENTION!

Before installing the bearings, make sure that the shaft mounted parts inside the bearings are in place before installation.

Since the bearing is a high precision component, it is important to avoid ingress of dust and foreign matter, and hammering during cleaning and installation. Use extreme care and insure clean conditions during installation and assembly.

ATTENTION!

The best way for bearing installation is heat shrinking. Knocking and hammering during installation should be avoided absolutely.

The bearing should be heated in a bath of clean oil at a temperature of approximately 80°C. After warming, slide the bearings in place quickly and nimbly so that it has not shrunk before being fully in position.

Grease the bearing after the temperature returns to normal, and then reassemble the motor.

5.6 Maintenance of sleeve bearings

5.6.1 Daily inspections

- (a) Ensure the volume and quality of lubrication oil are in compliance with specifications.

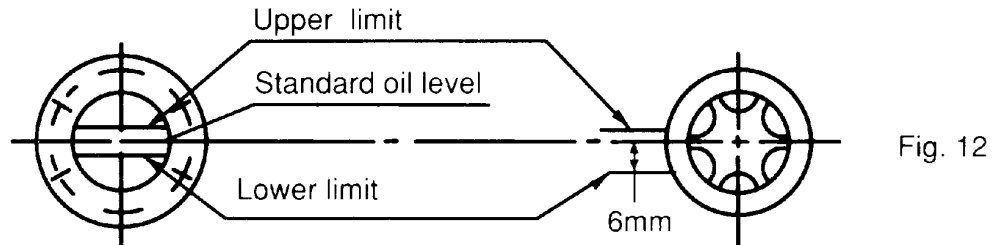


Fig. 12

- (b) Ensure there is motion of the oil ring and it is not clamped.
(c) The indicator of the shaft endplay should be restricted within the specified range of the red groove of the shaft or the $\pm 3\text{mm}$ (.118 in.) range of the drive-end shaft shoulder, or the bearing may be damaged.

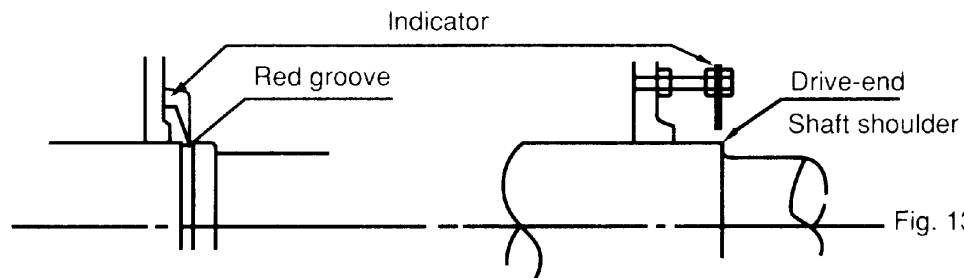


Fig. 13

5.6.2 Regular examination

- (a) Periodical change of oil

The oil reservoirs of self (not flood) lubricated bearings should be drained and refilled about every six (6) months. More frequent changes may be needed on high speed (3600 rpm) motors or if severe oil discoloration or contamination occurs. In conditions where contamination does occur, it may be advisable to flush the reservoir with kerosene to remove any sediment before new oil is added. Proper care must be taken to thoroughly drain the reservoir of the flushing material before refilling with the new oil.

Refill the reservoir to the center of oil sight glass with a rust and oxidation inhibited turbine grade oil. Refer to the outline and lubrication nameplate for the correct viscosity.

- (b) Quantity of lubrication oil

Please refer to the lubrication nameplate for oil quantity.

- (c) Oil viscosity

- (d)

ISO	Equivalents	Viscosity (SUS/100°F)
VG32	Esso Teresso 32	150
VG46	Esso Teresso 46	200
VG68	Esso Teresso 68	300

5.6.3 Disassembly



Prior to disassembling, ensure the power supplies are disconnected and there are no moving parts.

The bearing sleeve is of the spherically seated, self-aligning type. The opposite drive end bearing is normally insulated for larger motors (or when specified). On some motors, the insulation is bonded to the spherical seat of the bearing housing.

ATTENTION!

Extreme care must be exercised in removing the bearing sleeve from the insulated support to avoid damaging this insulation.

The following is the recommended procedure for removing the bearing sleeve:

- (a) Remove the oil drain plug in the housing bottom and drain the oil sump.
- (b) Remove all instrumentation sensors that are in contact with the bearing sleeve. These would include resistance temperature detectors, thermocouples, thermometers, etc..
- (c) Remove the socket head bolts holding the bearing cap and the inner air seal. The end cover plate must also be removed if the non-drive end bearing is being disassembled. Remove the bearing cap and top half of the inner air seal. Place them on a clean, dry surface to avoid damage to the parting surfaces.
- (d) Remove the top half of the bearing sleeve using suitable eyebolts in the tapped holes provided. Lift the bearing top straight up and avoid any contact with the shoulders of the shaft journals that might damage the thrust faces of the bearing. Place on a clean, dry surface taking care to prevent damage to either the parting surfaces or the locating pins that are captive in the top bearing half.
- (e) Remove the screws at the partings in the oil ring and dismantle the ring by gently tapping the dowel pin ends with a soft face mallet. Remove the ring halves and immediately reassemble them to avoid any mix up of parts or damage to the surfaces at the partings.
- (f) Pull up on the garter spring surrounding the floating labyrinth seal and carefully slip out the top half. Rotate the garter spring until the lock is visible. Twist counter-clockwise to disengage the lock, remove the garter spring then rotate the lower half of the seal out of the groove in the bearing housing. Note the condition of these floating labyrinth seals. If they are cracked or chipped, they must be replaced. Do not attempt to reuse a damaged seal.
- (g) To remove the bottom bearing half, the shaft must be raised a slight amount to relieve pressure on the bearing. On the drive end, this can be done by jacking or lifting on the shaft extension. Protect the shaft. On the non-drive, jacking or lifting can be done using bolts threaded into the tapped holes provided in the shaft end.

- (h) Roll the bottom bearing half to the top of the shaft journal and then lift it using suitable eyebolts threaded into the holes provided. Again avoid any contact with the shaft shoulders that could damage the bearing thrust faces. Place the lower bearing half on a clean, dry surface to protect the parting surfaces.



Use extreme care when rolling out the lower bearing half. Keep the hands and fingers well clear of any position where they might be caught by the bearing half if it were accidentally released and rotated back to its bottom position. Serious personal injury could result.

- (i) Protect the shaft journal by wrapping it with clean, heavy paper or cardboard.

5.6.4 Re-assembly

Bearing re-assembly is basically a reverse of the disassembly procedures outlined above, with the following suggestions:

- (a) The interior of the bearing housing should be cleaned and then flushed with clean oil or kerosene.
- (b) The bearing halves and the shaft journal should be wiped clean using lint-free cloth soaked with clean oil.
- (c) All parts should be carefully inspected for nicks, scratches, etc., in any contact surfaces. Such imperfections should be removed by an appropriate method such as stoning, scraping, filling, etc., followed by thorough cleaning.
- (d) Before installing the floating labyrinth seal halves, observe their condition. Do not attempt to use a cracked or chipped seal. The bottom half seal has a set of drilled holes in its side face. These must be placed at the bottom toward the inside of the bearing so that accumulating oil may drain back into the housing.
- (e) Put a bead of Curil-T around the seal half O.D.'s on both sides adjacent to the garter spring groove. This will prevent oil by-passing the seal around its outside.
- (f) Place the bottom seal half on top of the shaft and roll it into position. Install the top half and insert the garter spring pulling up on both ends to permit engaging the lock. Run a bead of Curil-T around the O.D's on both sides adjacent to the garter spring groove on this half also.
- (g) Carefully reassemble the two oil ring halves. Inspect the dowel pins for burrs and straightness and make any corrections required. Do not force the ring halves together. Excessive force may alter the roundness or flatness of the oil ring which can change its oil delivery performance.
- (h) Some of the pipe plugs in the housing are metric thread type. These are identified as those, which have a copper, lead, or similar material washer. If these plugs are removed,

be careful not to lose the washers. Before re-assembly, inspect the washers and replace them as required.

- (i) Before installing the bearing cap, observe the position of the floating labyrinth seal. The “tab” must be on top to engage the pocket. Failure to position the seal properly will result in damage when the cap is assembled.

ATTENTION!

- (1) Curil-T is the only approved compound for use in the assembly of the bearings on this motor. Other products may harden and impede the operation.**
- (2) During the re-assembly of the bearing parts, a thin layer of Curil-T should be applied to all gaskets and machined interface surfaces. This suggestion does not apply to the machined surfaces of the bearing liner halves.**
- (3) When seating the bearing shell, apply a thin layer of lube oil at the spherical surface of the liner. Slowly roll the lower bearing liner into the bearing housing making sure that the splined surface of the liner and the housing are flush. Gradually lower the shaft onto the bearing. The weight of the shaft will help rotate the bearing liner so that the babbitt surface of the liner will match the slope of the journal. Sometimes it is required to use a rubber mallet to tap lightly on the bearing housing while slowly rolling the shaft to help this seating operation.**

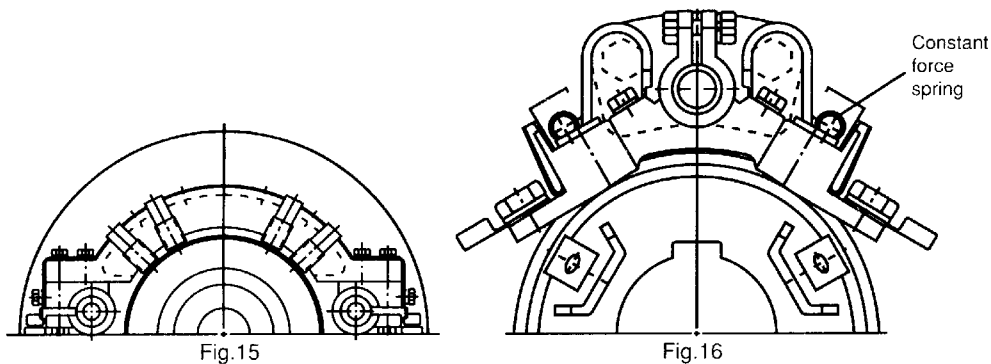
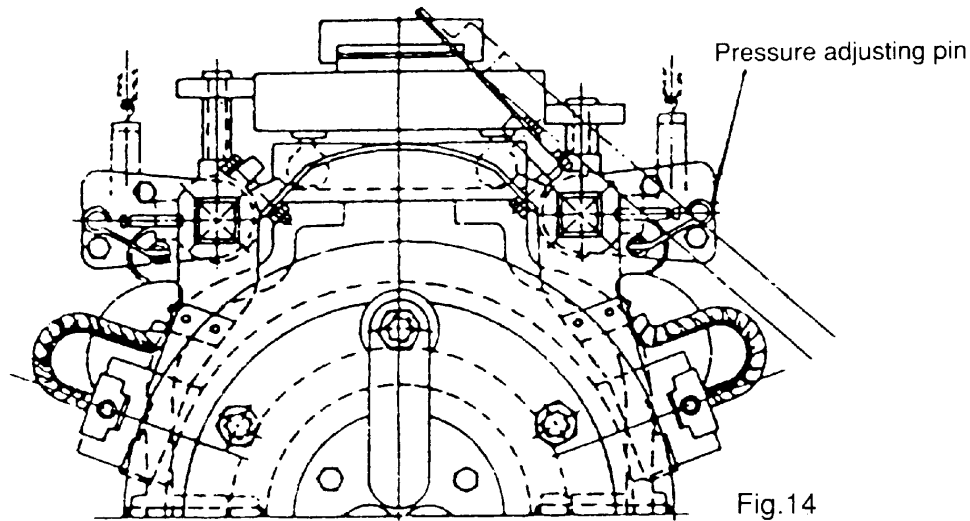
5.7 Maintenance of slip ring (For Wound Rotor Motors only)



Ensure motor is disconnected from power supplies and there are no accessible moving parts before maintenance operation.

5.7.1 Adjustment of carbon brush

- (a) Brush pressure for normal operation:
 - Electro-graphite brush.....0.2~0.25 kg/cm²
When frequent vibrations are evident or the brush is small (area below 0.5 cm²), the pressure should be greater than as shown.
- (b) Adjustment of brush pressure:
The brush pressure should be adjusted to keep normal operation as it wears.
 - The brush pressure may be reduced after use, so it is necessary to re-adjust. For adjustment, please turn adjusting screw, pressure adjusting pin or pressure adjusting plate as shown in Fig. 14 to obtain the correct tension ($=0.23 \times \text{brush cross sectional area in cm}^2$) $\pm 10\%$ kg.
- (c) Brush pressure need not be adjusted if constant force spring is used as shown in Fig. 15 and Fig. 16.



5.7.2 Brush replacement

The carbon brush is a part of the equipment which is easily worn away, replace it after it is worn to $\frac{1}{2} \sim \frac{3}{4}$ of original size.

(a) Brush material

The brush material is important to the performance of the motor. Only the most appropriate materials are chosen by TWMC, and are listed on the nameplate of the motor. It is important to know this when you replace the brush, so a recommended type is used.

(b) Dimensions

Brush, holder and gap between them, please refer to CNS 2322 C4051 or JIS C2802.

ATTENTION!

The gap between a brush and it holder is important for good performance and safety of the motor.

(c) Adjustment of new brushes (Shown in Fig. 17)

- (1) Polish the new brush with a file until it assumes the appropriate contour of the slip ring which it touches.

- (2) Place sand-paper (JIS R6252 No. 40...50) on the slip ring with the abrasive face of the paper against the brush to induce a closer contact by rubbing against each other.
- (3) Repeat item 2 with fine sand –paper (JIS R6252 No. 100 to 200) until the contact surface between brush and slip ring exceeds 80%.
- (4) Finally, clean the contaminated slip ring and brush with clean cloth or compressed air.

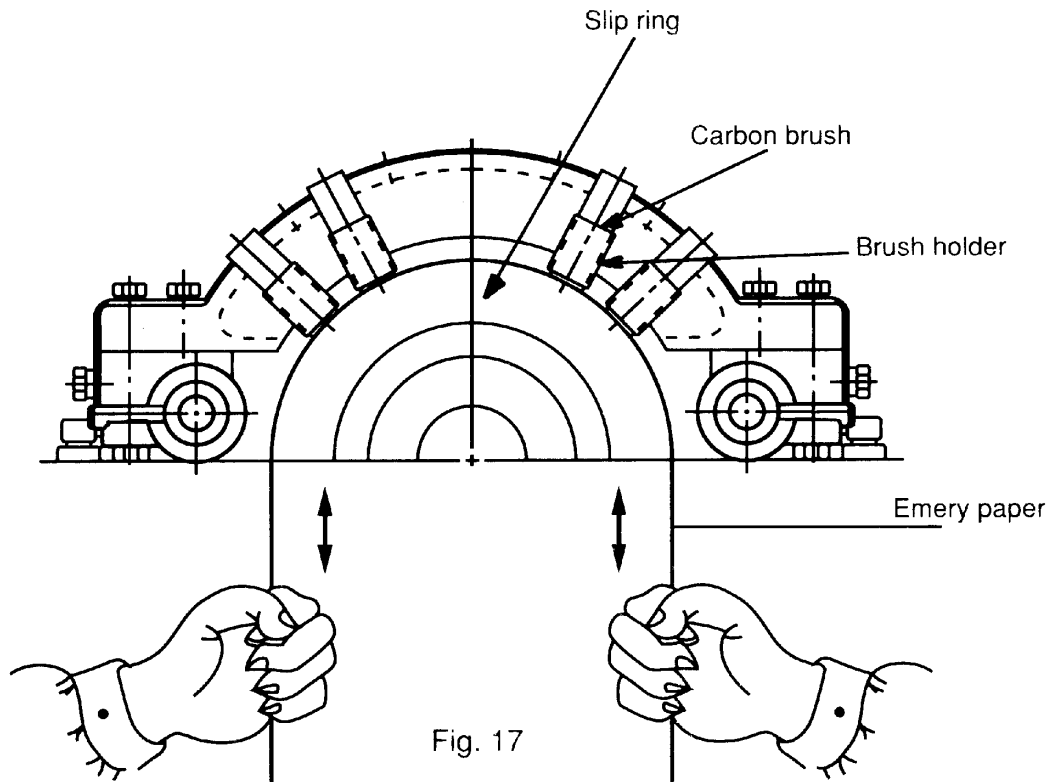


Fig. 17

5.8 Maintenance of non-reverse ratchet mechanism (For Vertical high Thrust Motor only)

5.8.1

In the pump piping system, a check valve and a stop valve should be installed in the discharge line. The check valve, placed between the pump and the stop valve, is to protect the pump from reverse flow and excessive backpressure. The stop valve is used in priming, starting and when shutting down the pump. It is advisable to close the stop valve before stopping the pump. This is especially important when the pump is operated against a high static head.

TWMC vertical high thrust motors are equipped with non-reverse ratchet (N.R.R.) mechanism only when requested by the pump manufacturer. Typical construction of the N.R.R. mechanism is shown as Fig 18 below.

The N.R.R. mechanism keeps the pump and motor from rotating in the reverse direction. Thus prevents damage from over-speeding and damage to water-lubricated pump shaft bearings

when, on shutdown, the falling water column tends to drive the pump in the reverse direction. In normal operation, the ratchet pins are lifted by the ratchet teeth and are held clear by centrifugal force and friction as the motor comes up to speed. When power is removed, the speed decreases and the pins fall. At the instant of reversal, a pin will catch in a ratchet tooth and prevent backward rotation.

5.8.2

The service life of ratchet pins depends not only on the reverse shock load between the pin and ratchet tooth when pump stopped but also the frequency of pump starting and stopping while in service. Provided that the pins are deformed due to this reverse shock load, then the up and down motion of the ratchet pins could be sluggish or jammed and that unusual noises shall arise.

The recommended replacement period for these ratchet pins is every three (3) years. If the reverse shock load is greater than 30% of motor rated torque or the starting frequency is more than twice per day, then the replacement period should be halved.

ATTENTION!
The check valve and stop valve in the discharge line should be regularly inspected and maintained to assure the normal function of these valves. This is important to protect the pump and motor from damage and increase the service life of the N.R.R. mechanism.

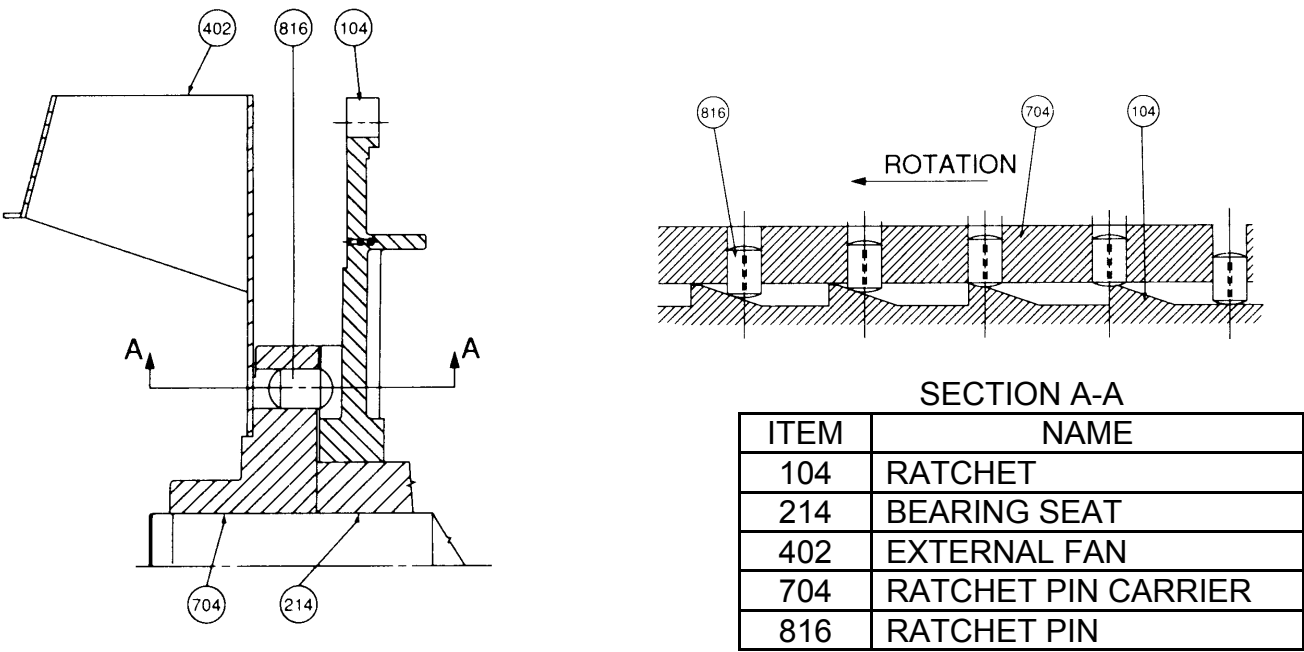


Fig. 18

6. FAULT FINDING AND RECOGNITION

Kinds of Breakdown	Symptoms	Possible Causes	Remedies
Fail to start without load	Motionless and soundless	Power-off	Consult power company
		Switch-off	Switch-on
		No fuse	Install fuse
		Broken wiring	Check wiring and repair
		Broken lead	Check wiring and repair
		Broken windings	Check windings and repair
	Fuse blowing. (Automatic switch trips off, slow start with electromagnetic noise)	Short circuit of circuit switches	Check circuit switches and replace
		Incorrect wiring	Check wiring according to nameplate
		Poor contact at terminal	Lock tightly
		Windings grounded	Factory repair
		Broken windings	Factory repair
		Poor contact of circuit switches	Check and repair
		Broken wiring	Check and repair
		Poor contact of starting switches	Check and repair
		Short circuit of starting switches	Check and repair
		Incorrect connections of starting switches	Connect according to nameplate
Loading after start	Fuse blowing. Fail to restart due to trip-off of automatic switch	Insufficient capacity of fuse	Replace fuse if wiring permits
		Overload	Lighten load
		High load at low voltage	Check circuit capacity and reduce load
	Overheating motor	Overload or intermittent overload	Lighten load
		Under-voltage	Check circuit capacity and power source
		Over-voltage	Check power source
		Ventilation duct clogged	Remove the foreign matter in the duct
		Ambient temperature exceeds 40°C	Correct insulation class to B or F, or lower ambient temperature
		Friction between rotor and stator	Factory repair
		Fuse blown (Single-phase rotating)	Install the specified fuse
		Poor contact of circuit switches	Check and repair
		Poor contact of circuit starting switches	Check and repair
		Unbalance three-phase voltage	Check circuit or consult power company

Kinds of Breakdown	Symptoms	Possible Causes	Remedies
Loading after start	Speed falls sharply	Voltage drop	Check circuit and powers source
		Sudden overload	Check machine
		Single-phase rotating	Check circuit and repair
	Switch overheat	Insufficient capacity of switch	Replace switch
		High load	Lighten load
	Bearing over-heating	High belt tension	Adjust belt tension
		Slack belt tension	Adjust belt tension
		Misalignment between motor and machine shafts	Re-align
		Over speed of bearing outer-ring	Adjust bracket
		High bearing noise	Replace the damaged bearing
Noise	Electromagnetic noise induced by electricity	Occurrence from its first operation	May be normal
		Sudden sharp noise and smoking	Short circuit of windings should be repaired at the factory
	Bearing noise	Noise of low shishi or Thru-Thru	May be normal
		Kala-Kala as a result of poor lubrication	Grease
		Kulo-Kulo as a result of poor lubrication	Clean bearing and grease
		Sa-Sa or larger noise	Replace the damaged bearing
	Mechanical noise caused by machinery	Loose belt sheave	Adjust key and lock the screw
		Loose coupling or skip	Adjust the position of couplings, lock key and screw
		Loose screw on fan cover	Lock fan cover screw tightly
		Fan rubbing	Adjust fan position
		Rubbing as a result of ingresson of foreign matter	Clean motor interior and ventilation ducts
		Wind noise	Noise induced by air flowing through ventilation ducts
		Induced by conveyance machine	Repair machine
Vibration	Electromagnetic vibration	Short circuit of winding	Factory repair
		Open circuit of rotor	Factory repair
	Mechanical vibration	Unbalanced rotor	Factory repair
		Unbalanced fan	Factory repair
		Broken fan blade	Replace fan
		Unsymmetric centers between belt sheaves	Align central points
		Central points of couplings do not lie on the same level	Adjust the central points of couplings to the same level
		Improper mounting installation	Lock the mounting screws
		Motor mounting bed is not strong enough	Reinforce mounting bed
		Remarks:	
(1) Circuit switches: These include knife switches, electromagnetic switches, fuse and other connection switches etc.			
(2) Starting switches: These include Delta-Star starters, compensate starters, reactance starters, resistor starters, starting controller’s etc.			